



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

Geometric & Topological Distances

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github.com/laplacebeltrami

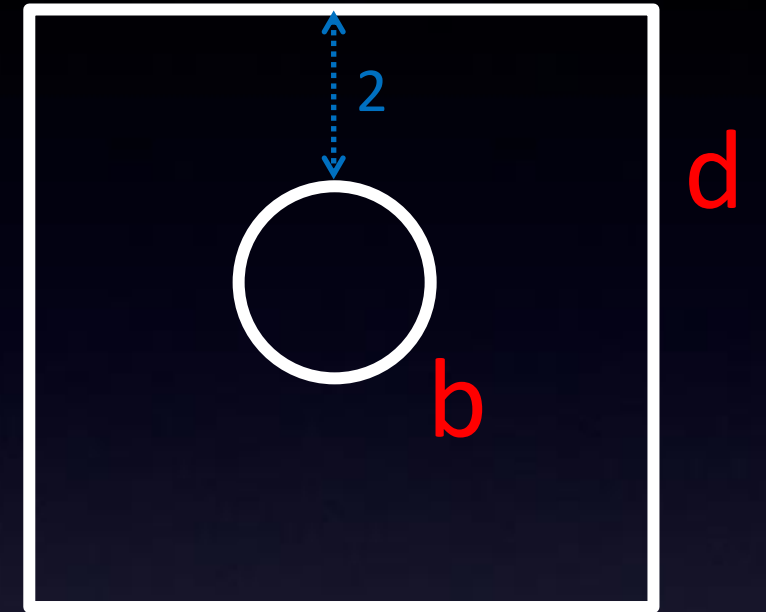
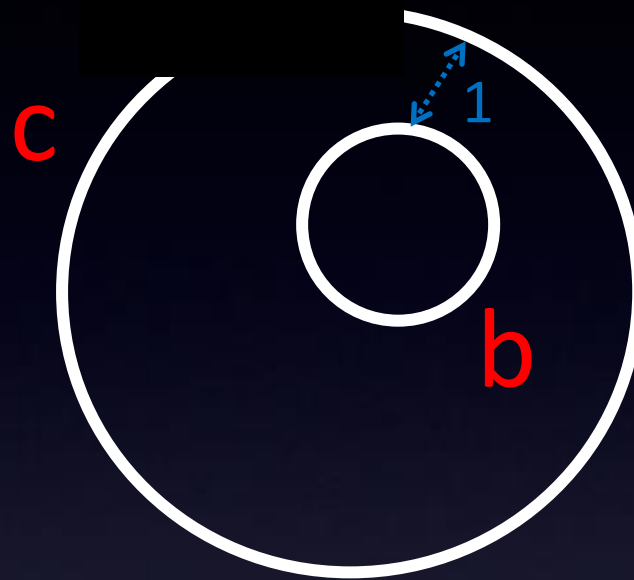
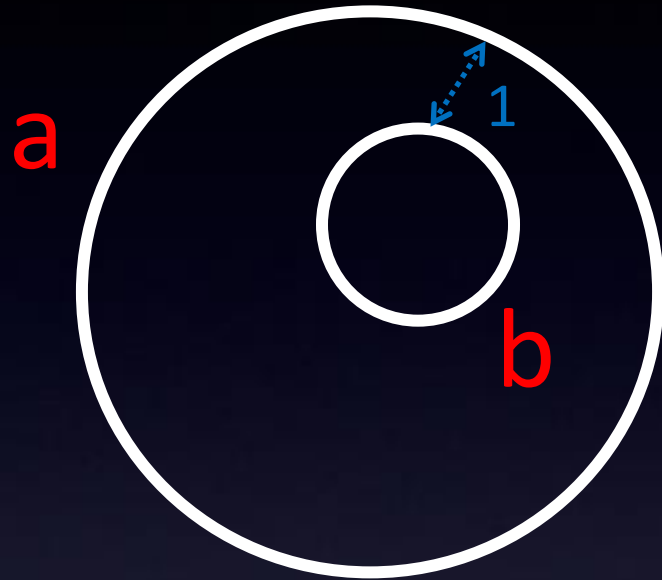
Learning Goals

1. Differentiate geometric and topological distance.
2. Can average by minimization
3. Understand Wasserstein distance

Geometric distance vs. Topological distances

d_{geo}

d_{top}



$$d_{geo}(a, b) = 1$$

$$d_{geo}(c, b) = 1$$

$$d_{geo}(d, b) = 2$$

$$d_{top}(a, b) = 0$$

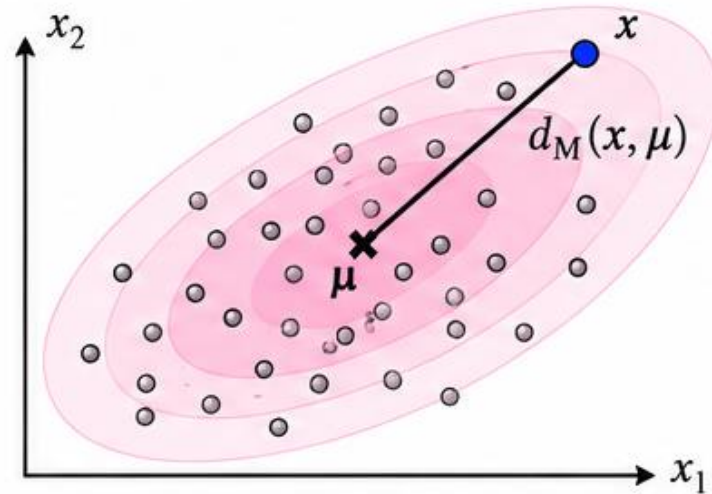
$$d_{top}(c, b) = 1$$

$$d_{top}(d, b) = 0$$

Example given in [*Chung et al. 2024 PLOS Computational*](#)

Mahalanobis distance

The Mahalanobis distance measures the distance between a point and a distribution while taking into account the covariance structure. It is the Euclidean distance after whitening (standardizing) by the covariance matrix.



Data with correlated variables
(elliptical distribution)

Definition

$$d_M(x, \mu) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

x : point ($d \times 1$)

μ : mean vector ($d \times 1$)

Σ : covariance matrix ($d \times d$, positive definite)

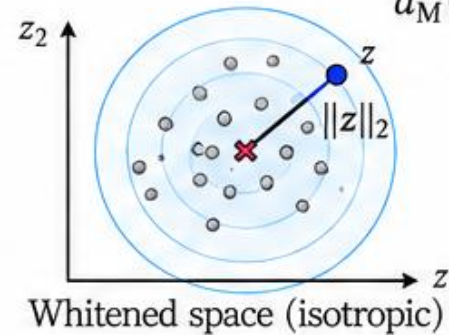
Geometric interpretation

Whitening transformation

$$z = \Sigma^{-1/2}(x - \mu) \longrightarrow$$

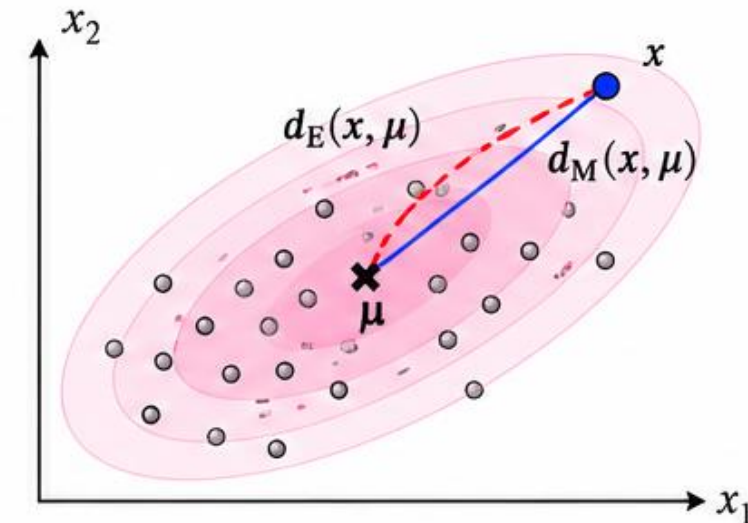
Mahalanobis distance
= Euclidean distance in
whitened space

$$d_M(x, \mu) = \|z\|_2$$



Whitened space (isotropic)

Comparison with Euclidean distance



--- Euclidean distance $d_E(x, \mu) = \|x - \mu\|_2$
— Mahalanobis distance $d_M(x, \mu)$

Mahalanobis distance accounts for the
covariance structure (shape, scale, and correlation).

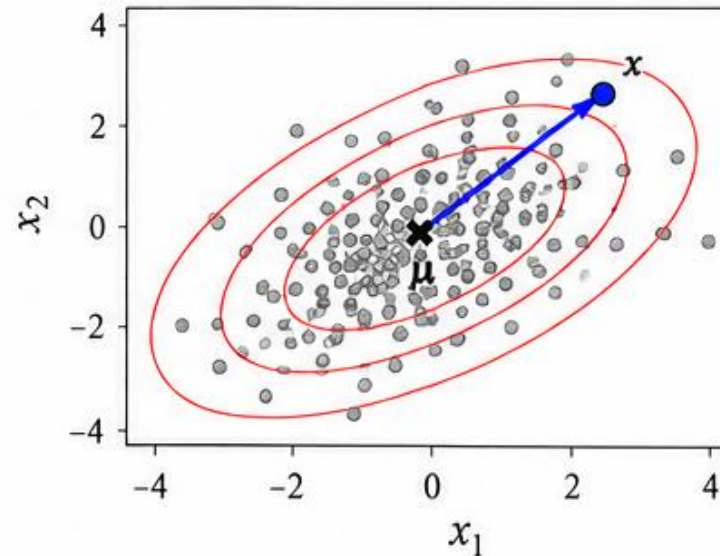
Matlab Implementation

```
% Generate correlated 2D data
n = 200;
mu = [0; 0];
Sigma = [3 1.5; 1.5 1];
X = mvnrnd(mu, Sigma, n);

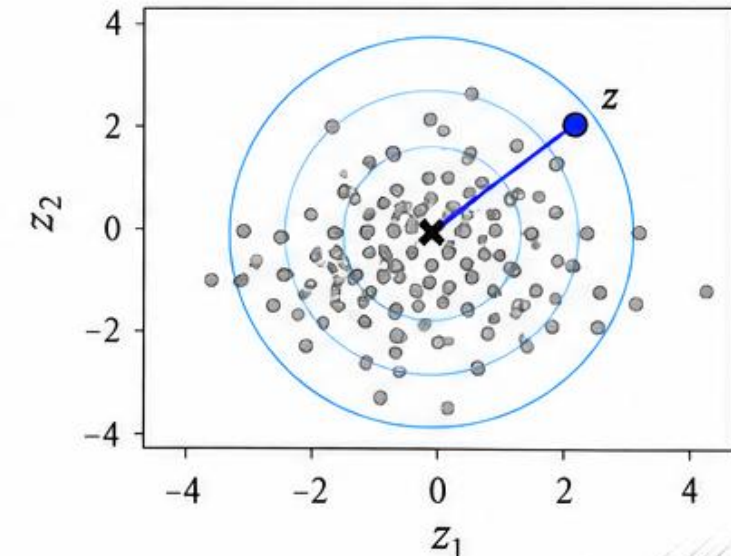
% Compute Mahalanobis distances
x0 = [3; 2];
Sigma_inv = inv(Sigma);
dM = sqrt(sum((X - x0') * Sigma_inv .* (X - x0'), 2));

% Plot
figure; hold on;
scatter(X(:,1), X(:,2), 30, 'filled', [0.6 0.6 0.6]);
contour_ellipse(mu, Sigma, [1 2 3], 'r');
plot(x0(1), x0(2), 'bo', 'MarkerSize', 8, 'MarkerFaceColor', 'b');
line([mu(1) x0(1)], [mu(2) x0(2)], 'Color', 'b', 'LineWidth', 2);
text(x0(1)+0.1, x0(2)+0.1, 'x'); text(mu(1)-0.2, mu(2)-0.2, '\mu');
axis equal; xlabel('x_1'); ylabel('x_2');
title('Mahalanobis distance');
```

Example: correlated Gaussian data



Same data in whitened space



Remark: Mahalanobis distance can be viewed as distance along manifold

Fréchet Mean vs. Karcher Mean

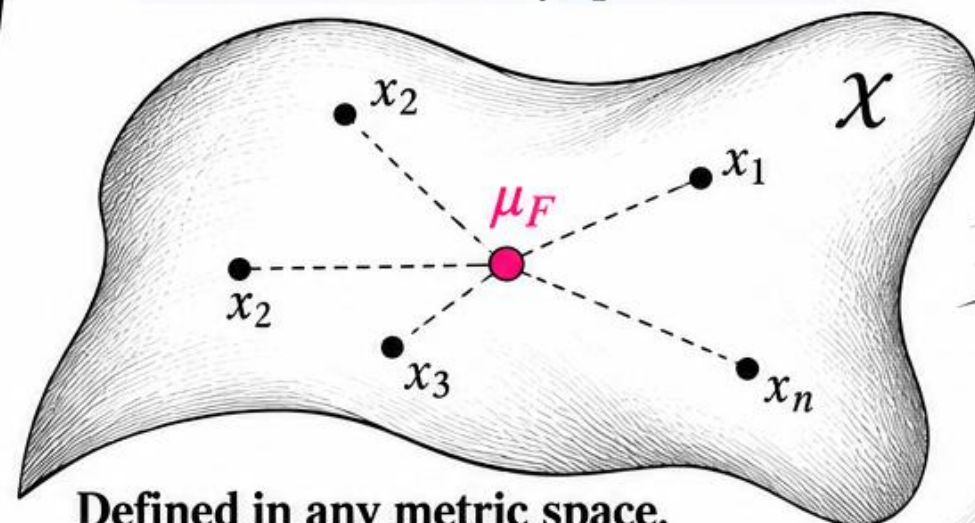
Same idea. Different mathematical settings.

A notion of average
in a curved world.

Fréchet mean (general)

Metric space $(\mathcal{X}, d)$

$$\mu_F = \arg \min_{x \in \mathcal{X}} \sum_{i=1}^n d^2(x, x_i)$$



Defined in any metric space.

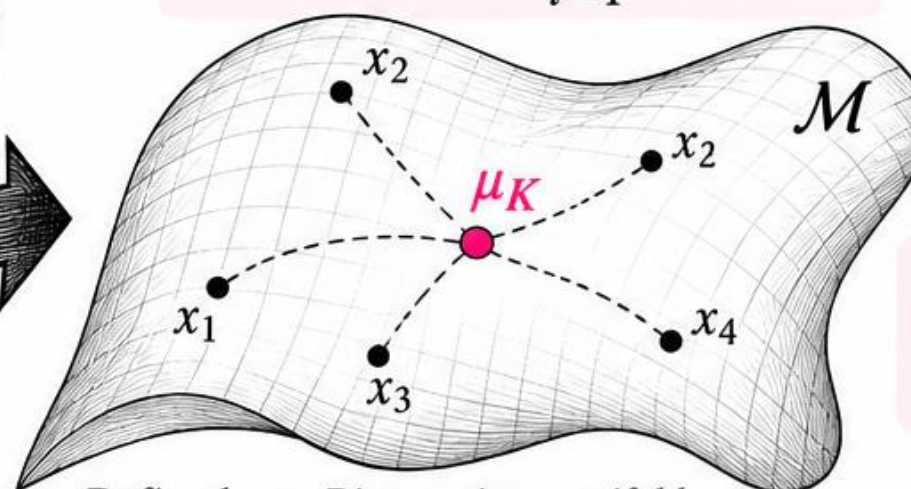
Uses the metric distance d .

May have multiple solutions or none.

Karcher mean (Riemannian)

Riemannian manifold $(\mathcal{M}, g)$

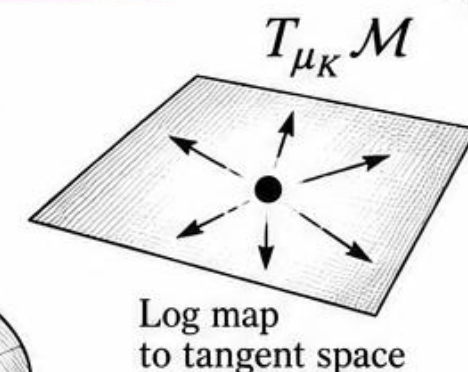
$$\mu_K = \arg \min_{x \in \mathcal{M}} \sum_{i=1}^n d_g^2(x, x_i)$$



Defined on a Riemannian manifold.

Uses the geodesic distance d_g induced by g .

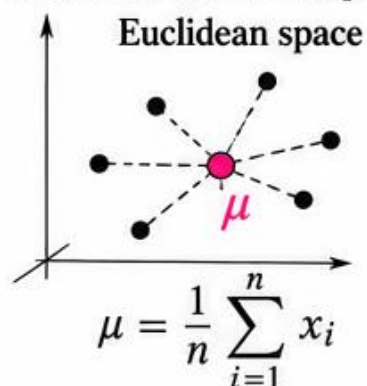
Often unique locally (e.g., within a convex neighborhood).



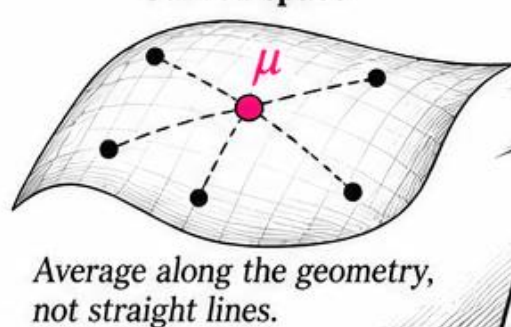
$$\mu_K = \text{Exp}_{\mu_K} \left(\frac{1}{n} \sum_{i=1}^n \log_{\mu_K}(x_i) \right)$$

Same intuition

Minimize summed squared distances to find an average.



Curved space



Key relationship

The Karcher mean is the Fréchet mean with the geodesic distance on a Riemannian manifold.

$$\mu_K = \mu_F$$

(when $\mathcal{X} = \mathcal{M}$ and $d = d_g$)

Same idea.
Different spaces.
A meaningful
average.

2-Wasserstein distance between scatter points

Random variables: $X \sim f_1 \quad Y \sim f_2$

2-Wasserstein distance: $\mathcal{D}(X, Y) = \left(\inf \mathbb{E} \|X - Y\|^2 \right)^{1/2}$

Scatter points

$$P_1 = \{x_1, \dots, x_q\} \subset \mathbb{R}^2 \quad P_2 = \{y_1, \dots, y_q\} \in \mathbb{R}^2$$

Empirical distributions

$$f_1(x) = \frac{1}{q} \sum_{i=1}^q \delta(x - x_i) \quad f_2(y) = \frac{1}{q} \sum_{i=1}^q \delta(y - y_i)$$

$$\mathcal{L}(P_1, P_2) = \inf_{\psi: P_1 \rightarrow P_2} \left(\sum_{x \in P_1} \|x - \psi(x)\|^2 \right)^{1/2}$$


Optimal
bijection

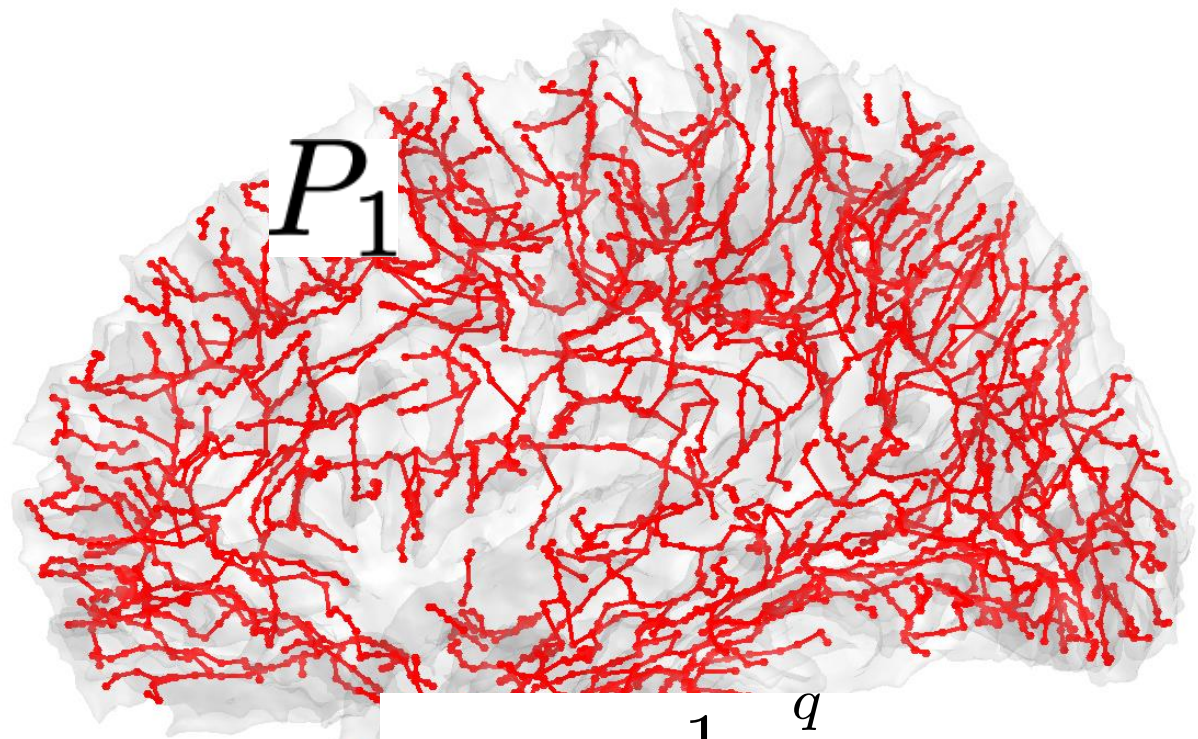
Assignment problem: Hungarian algorithm
 $\mathcal{O}(q^3)$

Dual convex formulation $\mathcal{O}(1)$

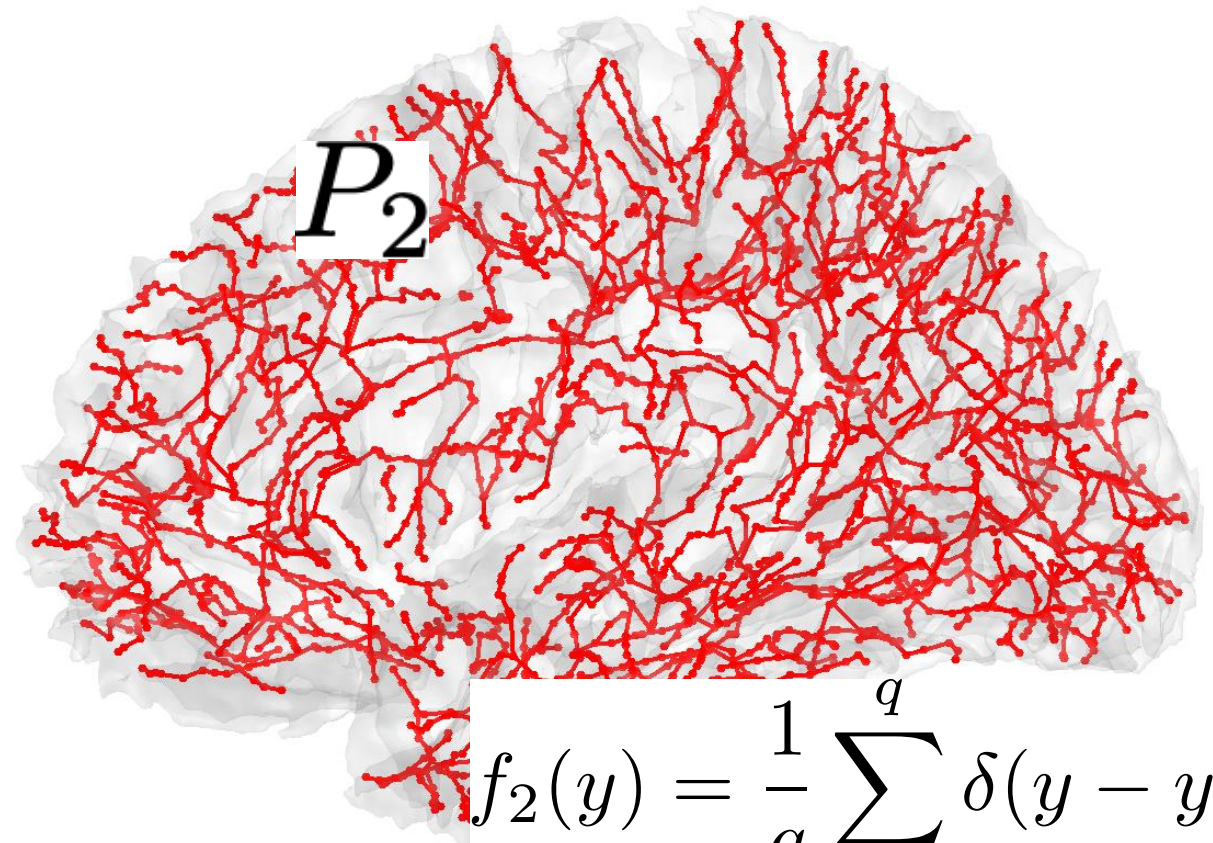
Registration of sucral patterns via Wasserstein distance

Random variables: $X \sim f_1$ $Y \sim f_2$

2-Wasserstein distance: $d(X, Y) = \left(\inf_{f(X, Y)} \mathbb{E} \|X - Y\|^2 \right)^{1/2}$



$$f_1(x) = \frac{1}{q} \sum_{i=1}^q \delta(x - x_i)$$



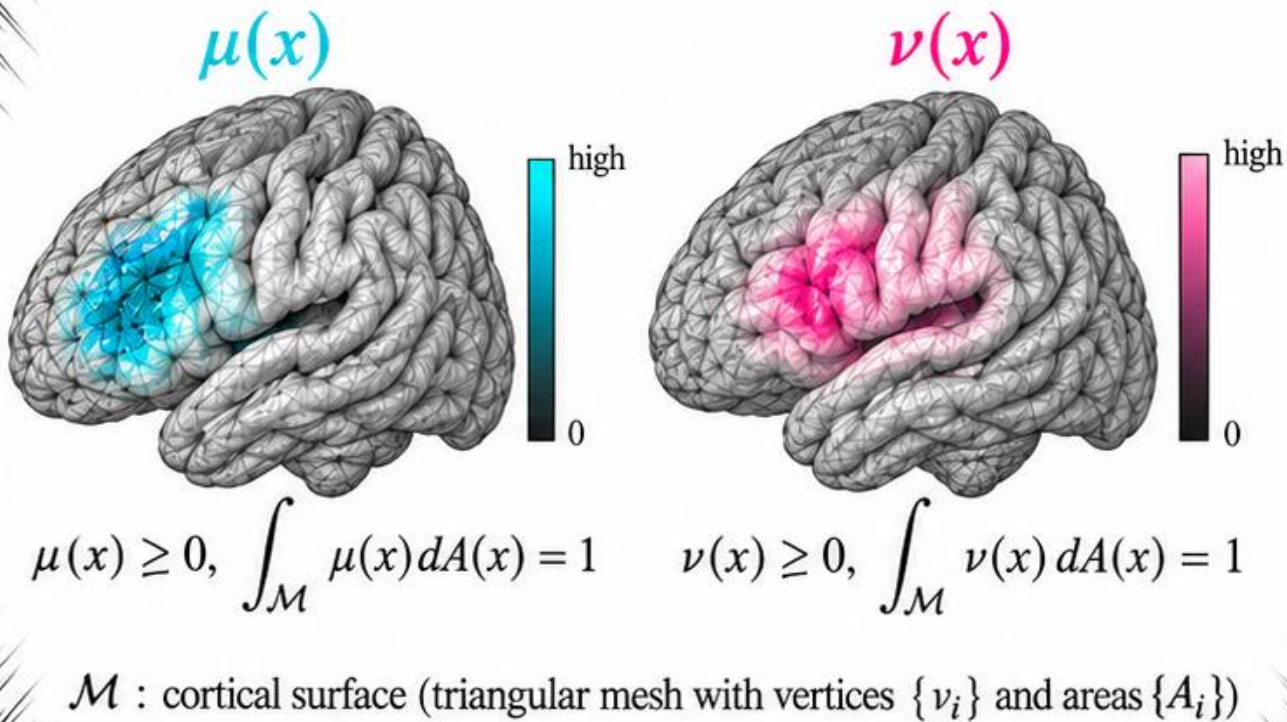
$$f_2(y) = \frac{1}{q} \sum_{i=1}^q \delta(y - y_i)$$

Finding optimal permutation ψ :

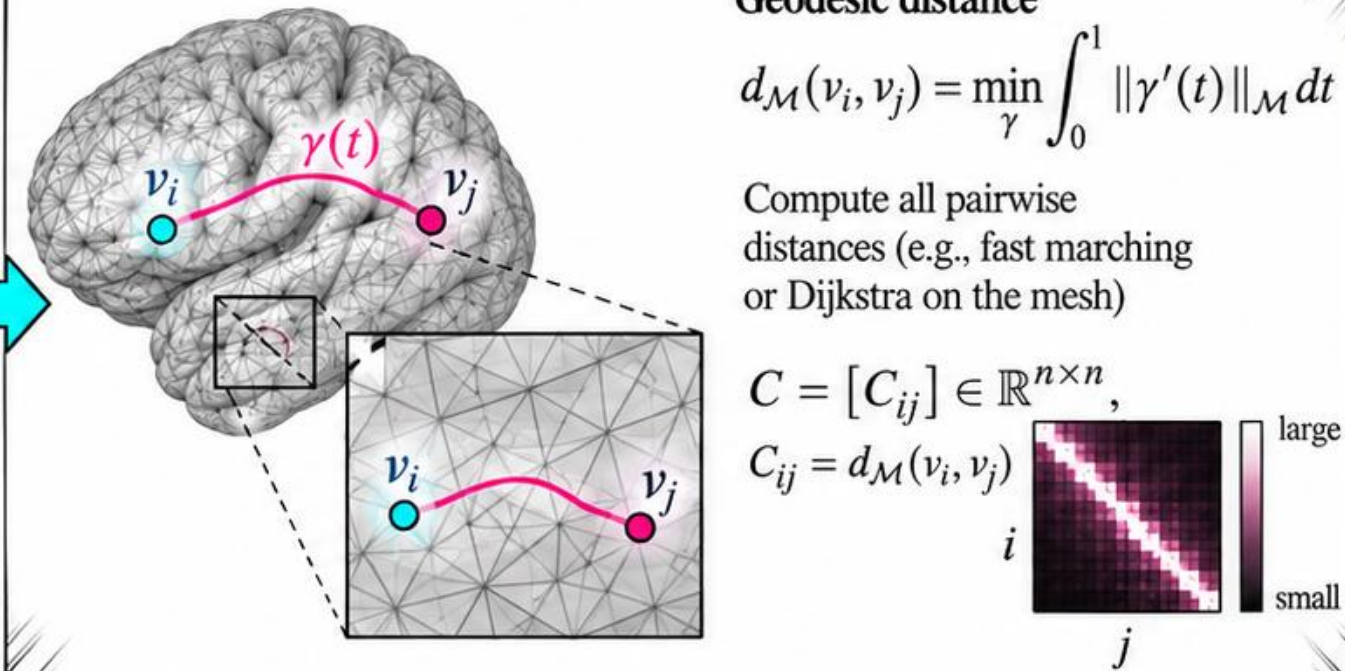
$$d(P_1, P_2) = \inf_{\psi: P_1 \rightarrow P_2} \left(\sum_{x \in P_1} \|x - \psi(x)\|^2 \right)^{1/2}$$

Wasserstein distance on manifolds data

1. Two probability distributions on a surface $\mathcal{M}$



2. Compute geodesic distance matrix on $\mathcal{M}$

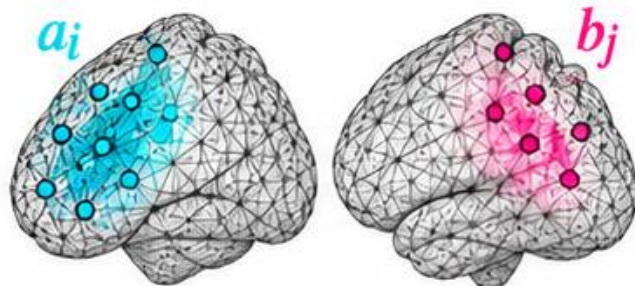


3. Discretize distributions and solve optimal transport

(a) Discretize μ and ν on the mesh

$$a_i = \mu(v_i)A_i, \quad b_j = \nu(v_j)A_j$$

$$a_i \geq 0, \quad \sum_{i=1}^n a_i = 1 \quad b_j \geq 0, \quad \sum_{j=1}^n b_j = 1$$



$\{v_i, A_i\}$ from the same mesh as in Step 2

(b) Solve the optimal transport problem

$$\min_{P \geq 0} \sum_{i,j} C_{ij} P_{ij}$$

$$\text{s.t.} \quad \sum_j P_{ij} = a_i \quad (i = 1, \dots, n)$$

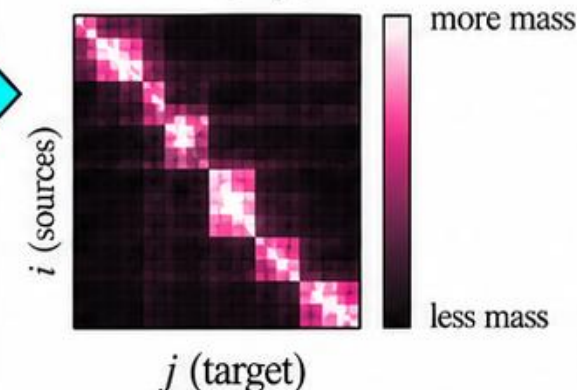
$$\sum_i P_{ij} = b_j \quad (j = 1, \dots, n)$$

$P = [P_{ij}]$: optimal transport plan
 (use Sinkhorn or linear programming)

4. Optimal transport map and 2-Wasserstein distance

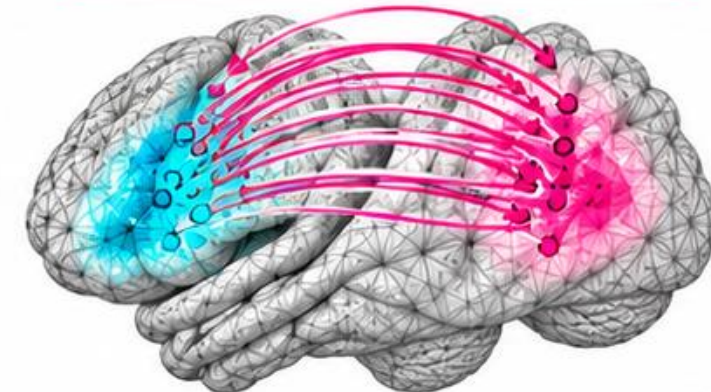
Transport plan

$$P = [P_{ij}]$$



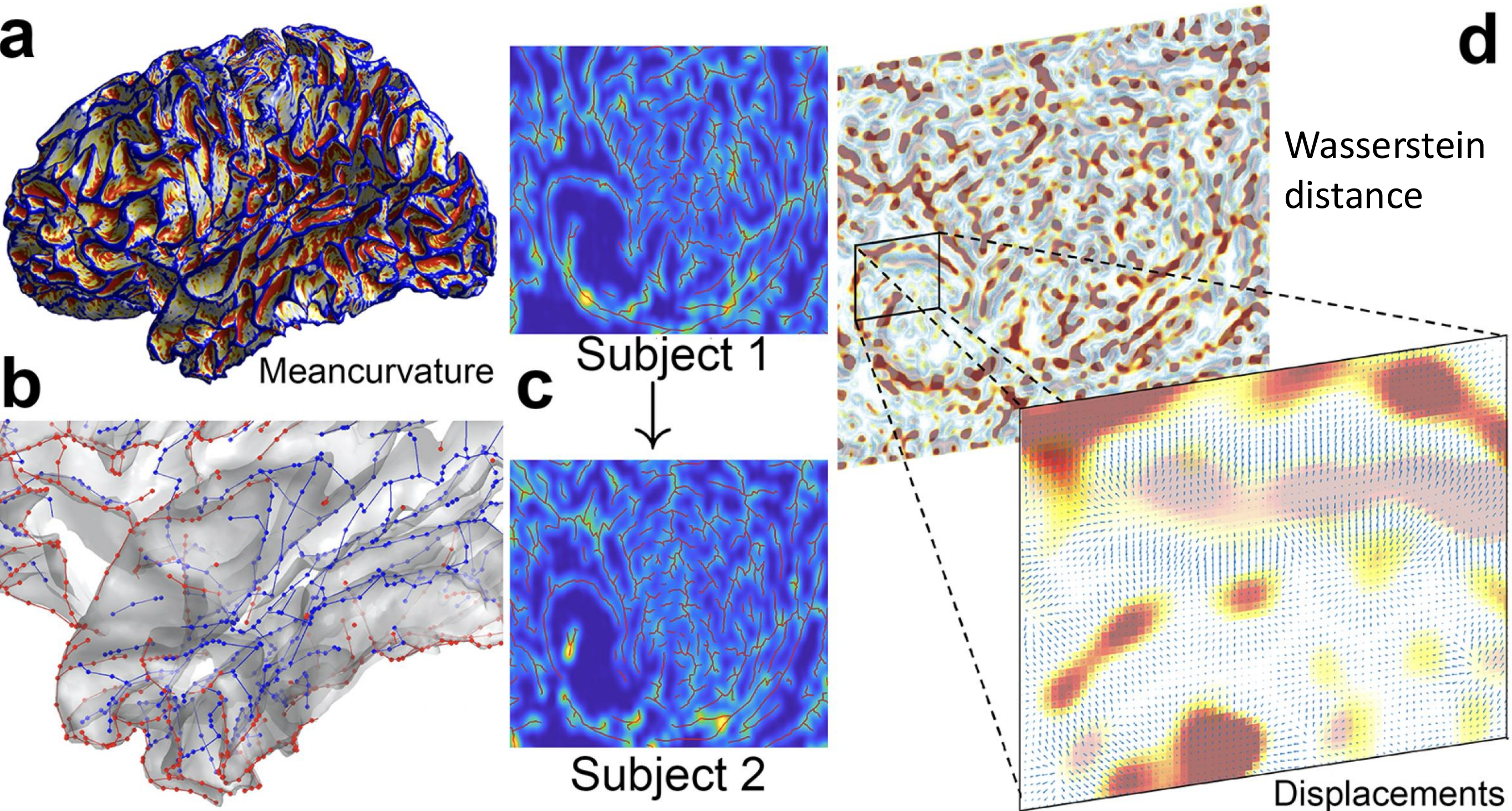
2-Wasserstein distance

$$W_2(\mu, \nu) = \left(\sum_{i,j} C_{ij} P_{ij} \right)^{1/2}$$



Project: build manifold version of Wasserstein distance

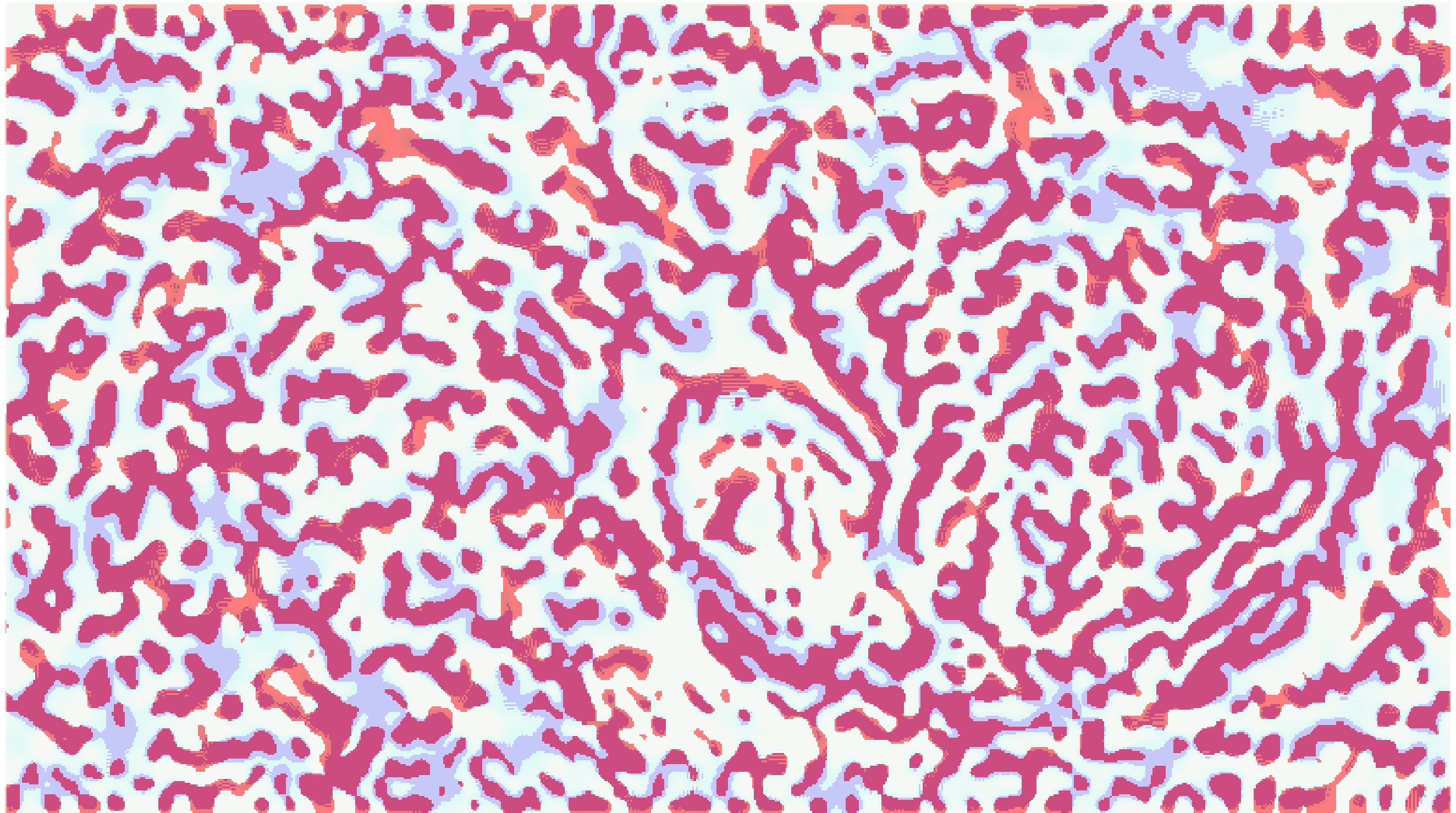
Registration of sulcal/gyral patterns



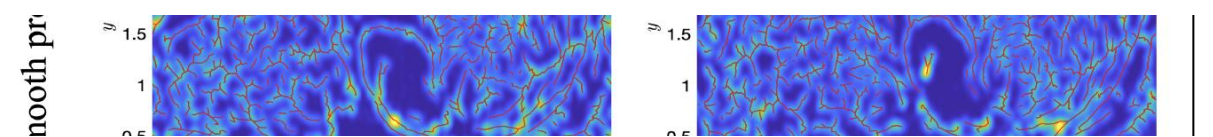
Huang et al. 2020 IEEE Transactions on Medical Imaging

Chen et al. 2023 arXiv:2307.00385

Wasserstein distance = Optimal transport on probability



performance. This method is applied in identifying the differences between male and female sulcal patterns.



1 INTRODUCTION

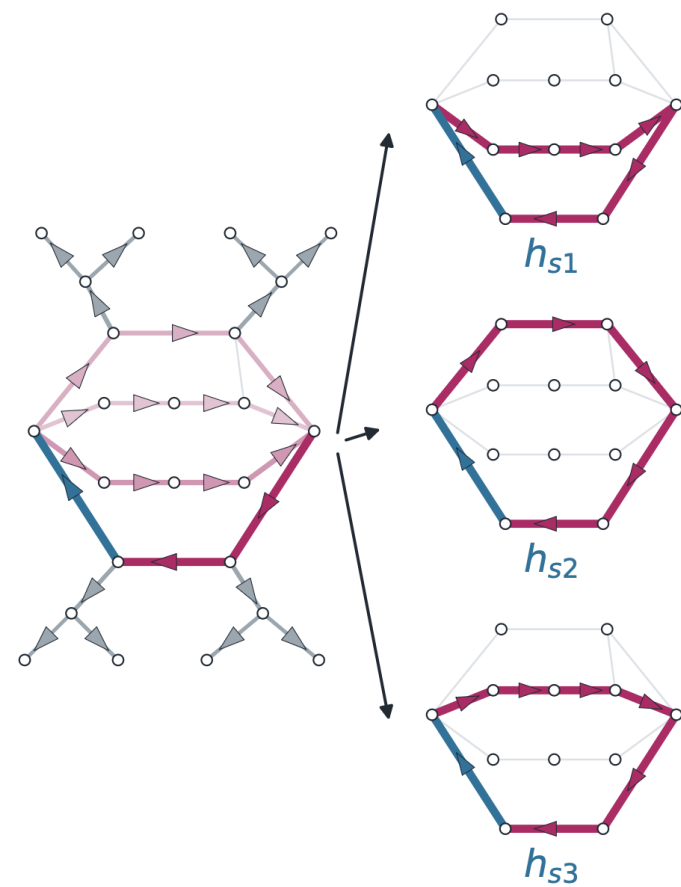
Codes:

<https://github.com/laplcebeltrami/sulc>

Chen et al. 2023

We can align many different nonstandard objects

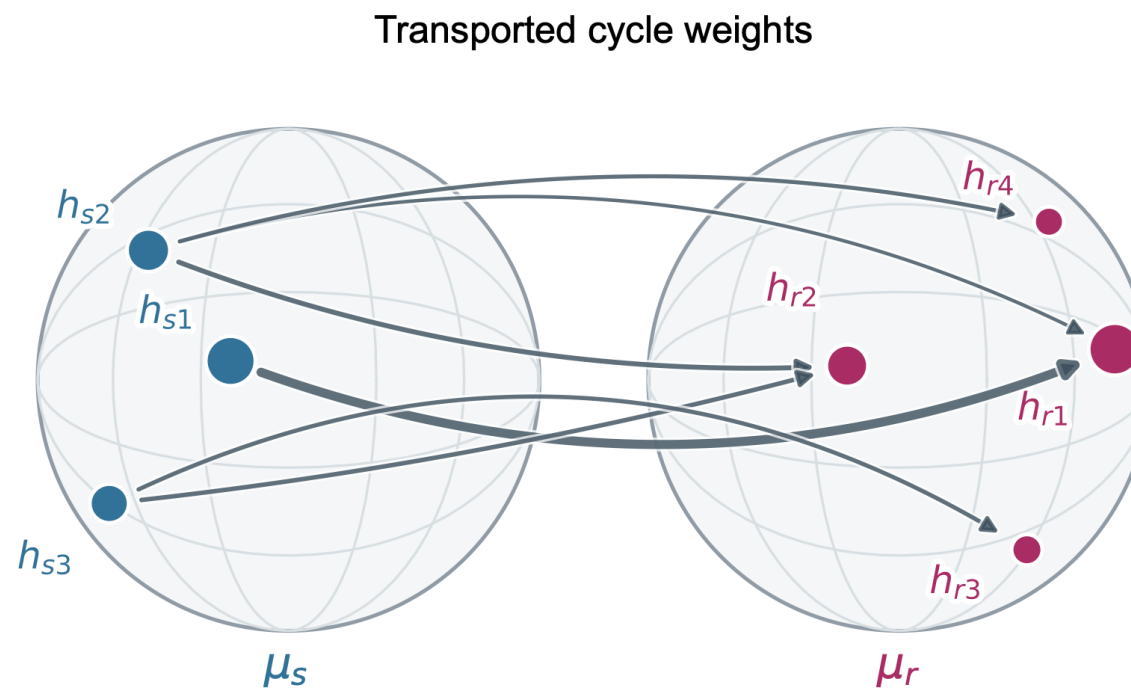
A. Source: 3 cycles



$$h_s = P_{\mathcal{H}} x_s$$

Project and extract

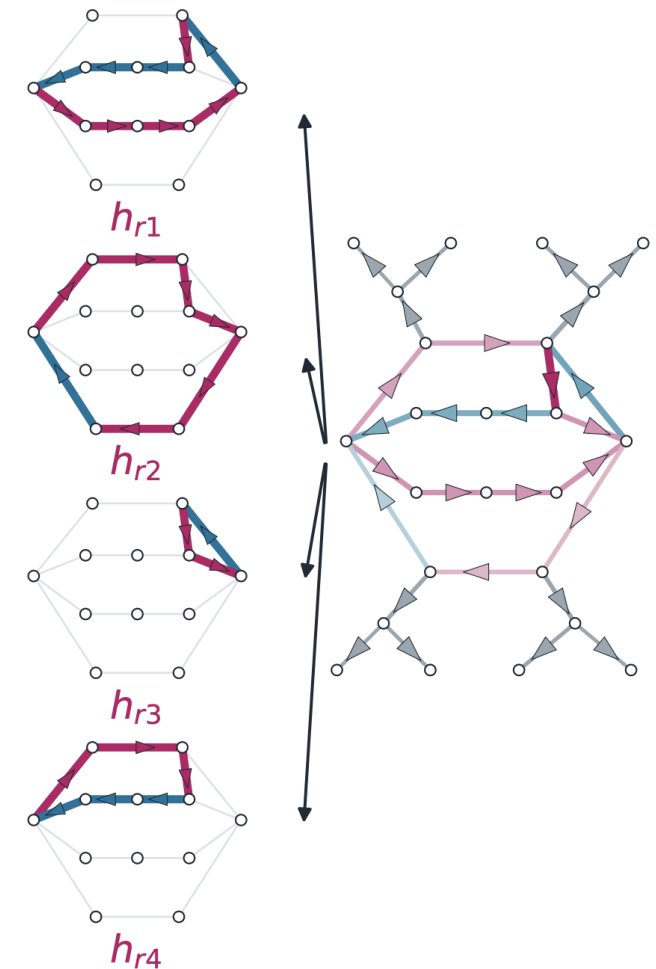
B. DC-Wasserstein alignment



Coupling π

	h_{r1}	h_{r2}	h_{r3}	h_{r4}
h_{s1}	0.45	0	0	0
h_{s2}	0.06	0.16	0	0.09
h_{s3}	0	0.13	0.11	0

C. Target: 4 cycles



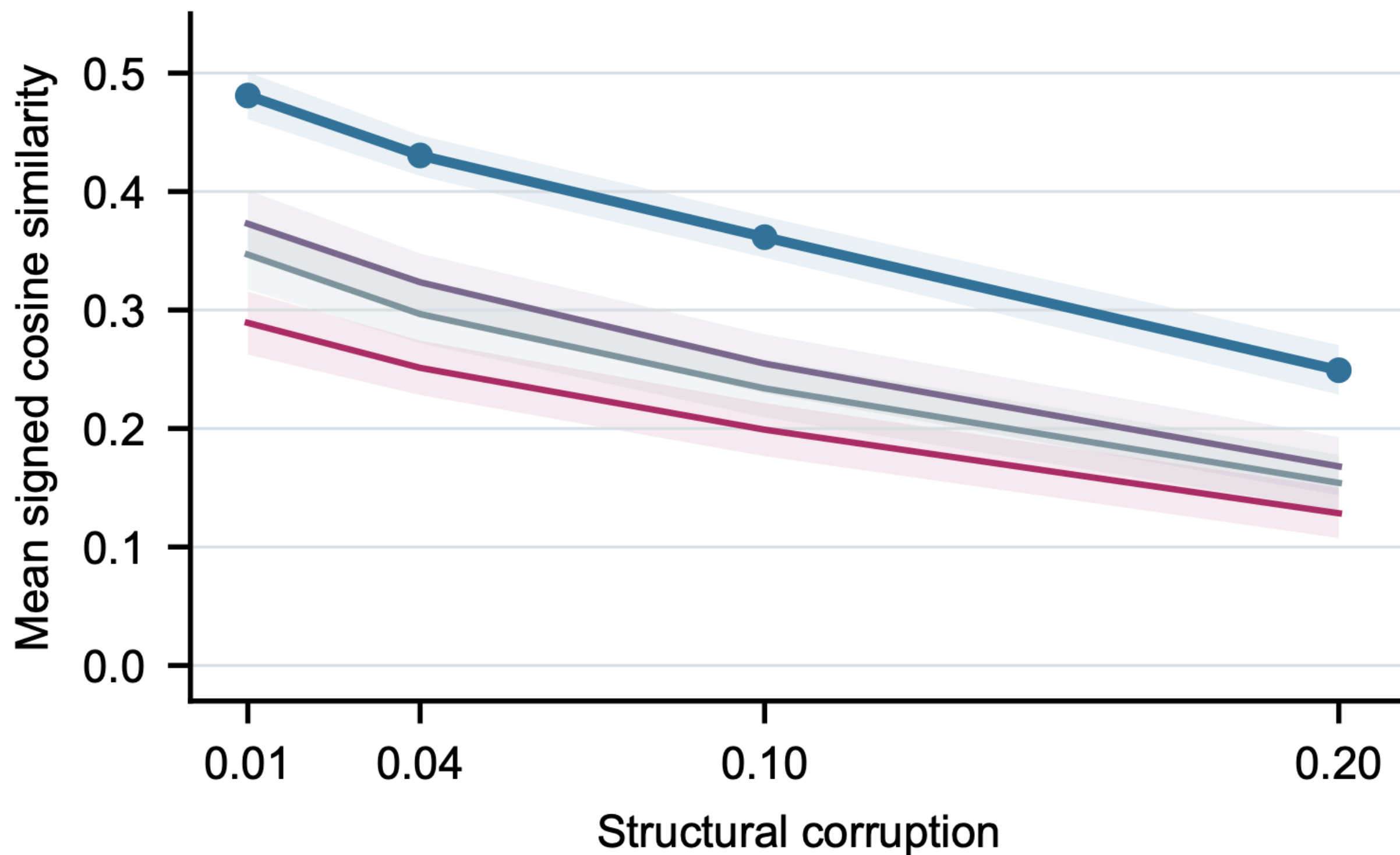
$$h_r = P_{\mathcal{H}} x_r$$

Project and extract

Figure 1: Many-to-many alignment of directed cycles. **A.** Projection extracts the source network's cyclic flow, which is decomposed into three directed cycles. **C.** The target contains four cycles with different routes and lengths. **B.** Each cycle is mapped to a point on a common cycle sphere; larger points indicate greater cycle weights α_{si} in Eq. (7). DC-Wasserstein transport aligns the cycles, allowing many-to-many correspondence.

Project: Align and register collection of cycles

All 2,160 alignment comparisons



- DC-Wasserstein
- Unoriented transport
- Image-persistence adapter
- TpOT incidence adapter
- Cosine ceiling

Topological Distance

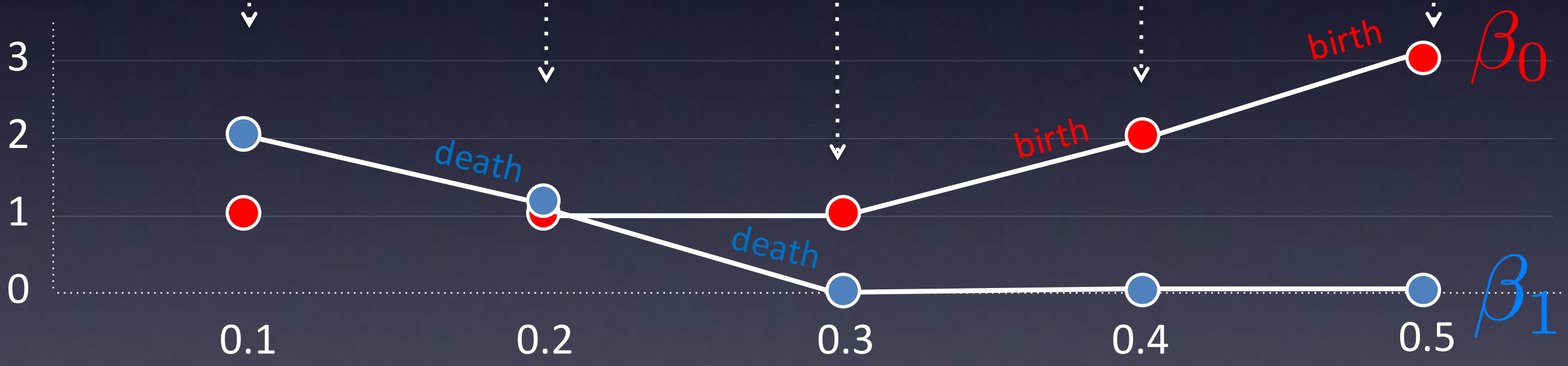
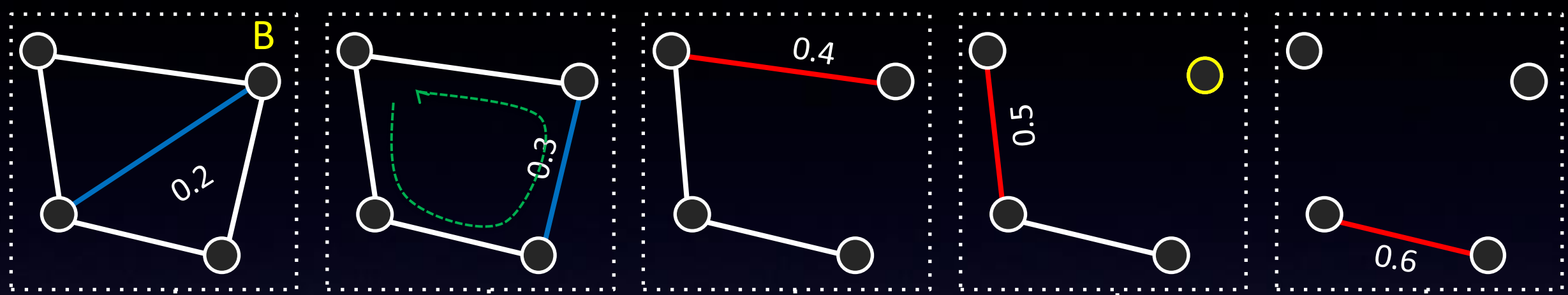
Required reading:

Songdechakraiwt and Chung. 2023, Annals of Applied

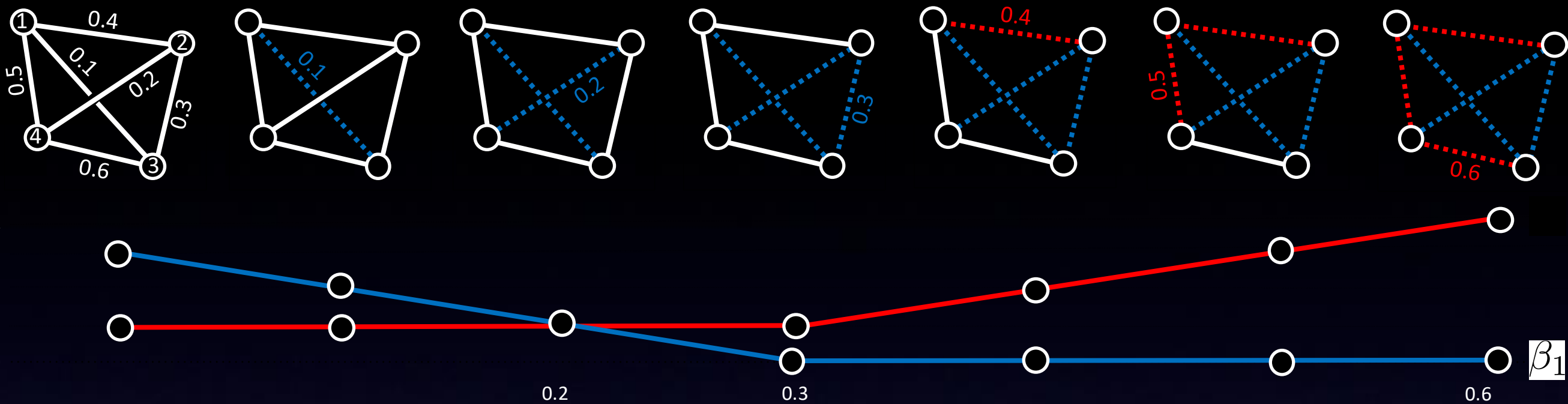
Persistence = Life time (death – birth) of a feature

Deleting edges destroy cycles

Deleting edges create components

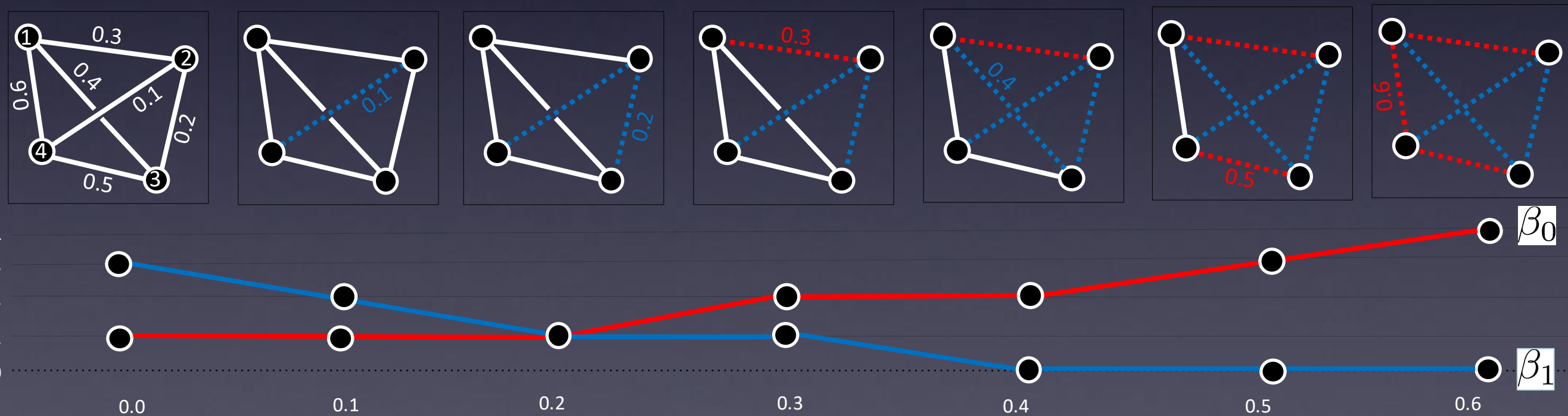


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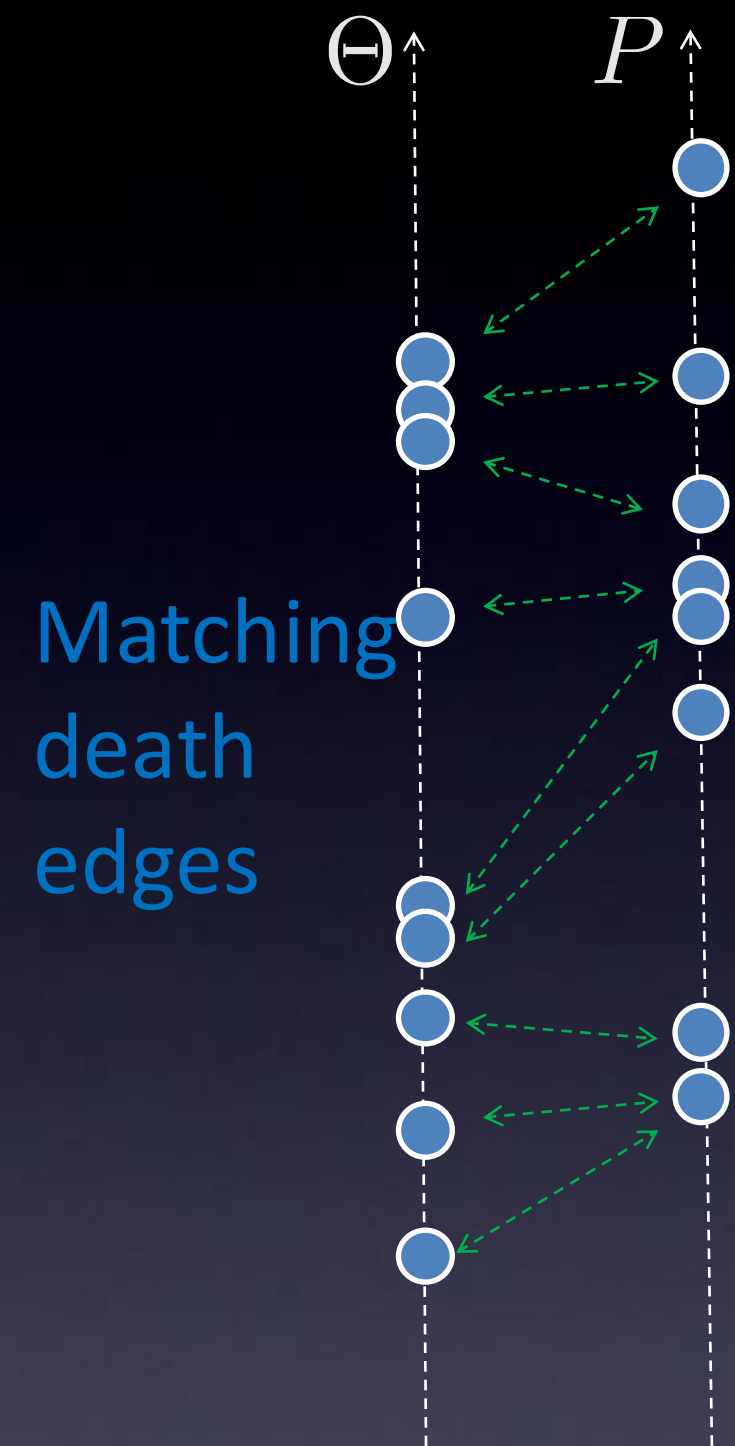


Topological distance computed using Wasserstein distance

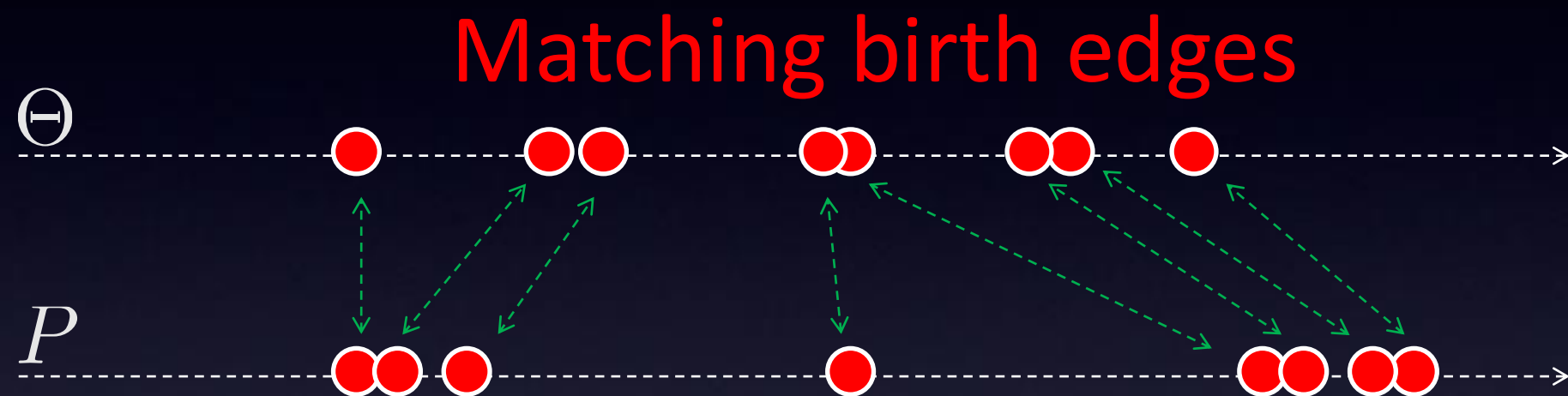
P



Topological distance via r -Wasserstein distance



$$d(X, Y) = \left(\inf_{f(X, Y)} \mathbb{E} \|X - Y\|^r \right)^{1/r}$$



$$d_0(\Theta, P) = \min_{\tau} \sum_b |b - \tau(b)|^r$$

Runtime $\mathcal{O}(q \log q)$

$$d_1(\Theta, P) = \min_{\tau} \sum_d |d - \tau(d)|^r$$

*Proof in
Chung et al. 2024,
Foundations*

Theorem: 2-Wasserstein distance (optimal transport) on graph filtrations

$$\begin{aligned}\mathcal{L}_{0D}(\Theta, P) &= \min_{\tau} \sum_{b \in E_0} [b - \tau(b)]^2 \\ &= \sum_{b \in E_0} [b - \tau_0^*(b)]^2\end{aligned}$$

Birth set

τ_0^* : The i -th smallest birth value to the i -th smallest birth value

$$\begin{aligned}\mathcal{L}_{1D}(\Theta, P) &= \min_{\tau} \sum_{d \in E_1} [d - \tau(d)]^2 \\ &= \sum_{d \in E_1} [d - \tau_1^*(d)]^2\end{aligned}$$

Death set

τ_1^* : The i -th smallest death value to the i -th smallest death value

Theorem: ∞ -Wasserstein distance on graph
filtrations

0D topology

$$\mathcal{L}_{0D}(\Theta, P) = \max_{b \in E_0} |b - \tau_0^*(b)|$$

Birth set

τ_0^* : The i -th smallest birth value to the i -th smallest birth value

1D topology

$$\mathcal{L}_{1D}(\Theta, P) = \max_{d \in E_1} |d - \tau_1^*(d)|$$

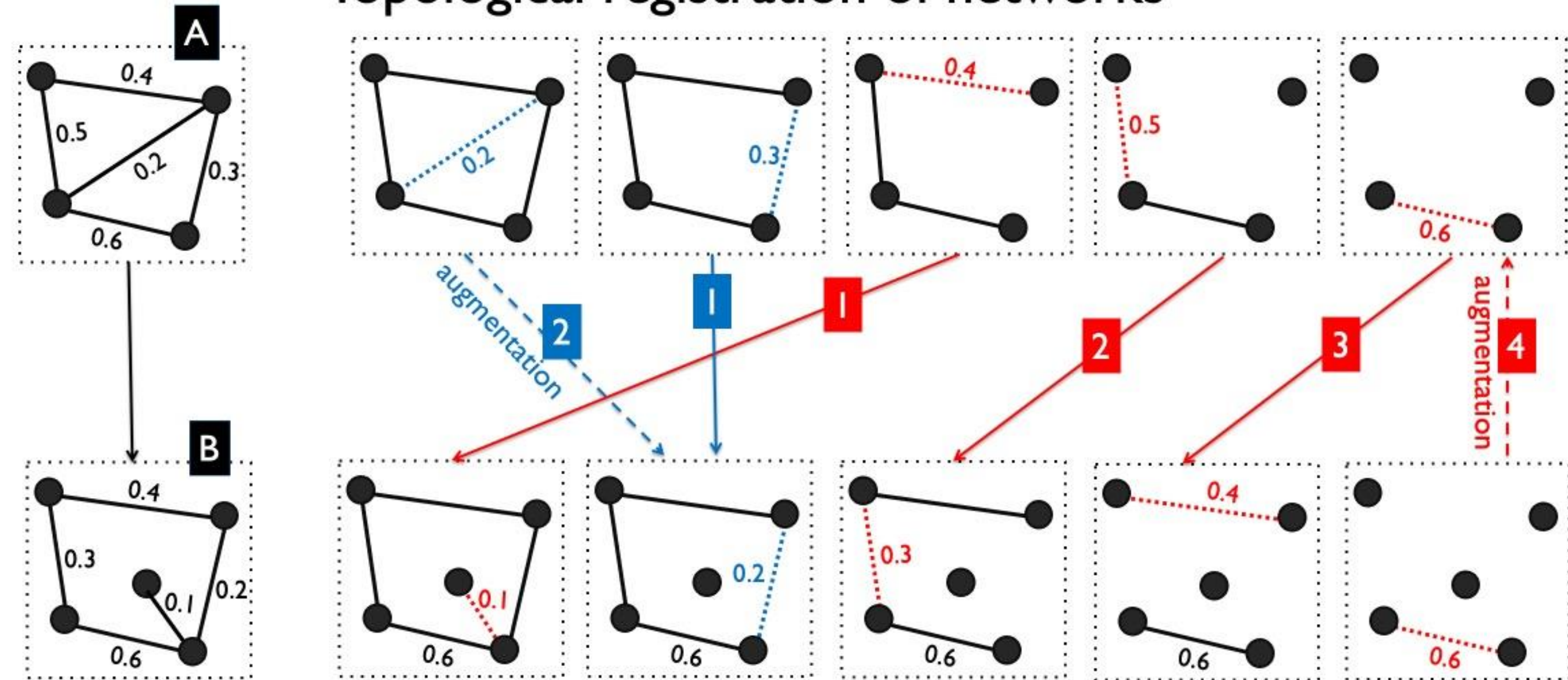
Death set

τ_1^* : The i -th smallest death value to the i -th smallest death value

Topological registration

→ needed to average topologically different objects

Topological registration of networks



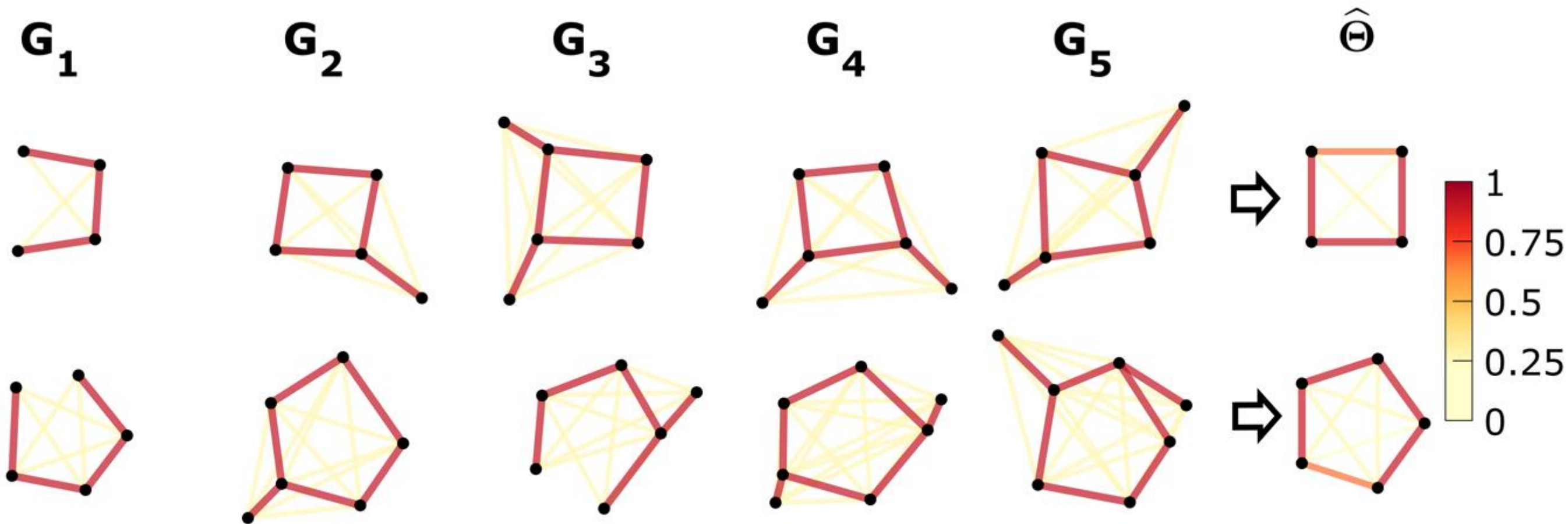
1) Match MST to MST, none-MST to none-MST

2) Match largest persistence (death – birth) first

Topological registration of topologically different objects

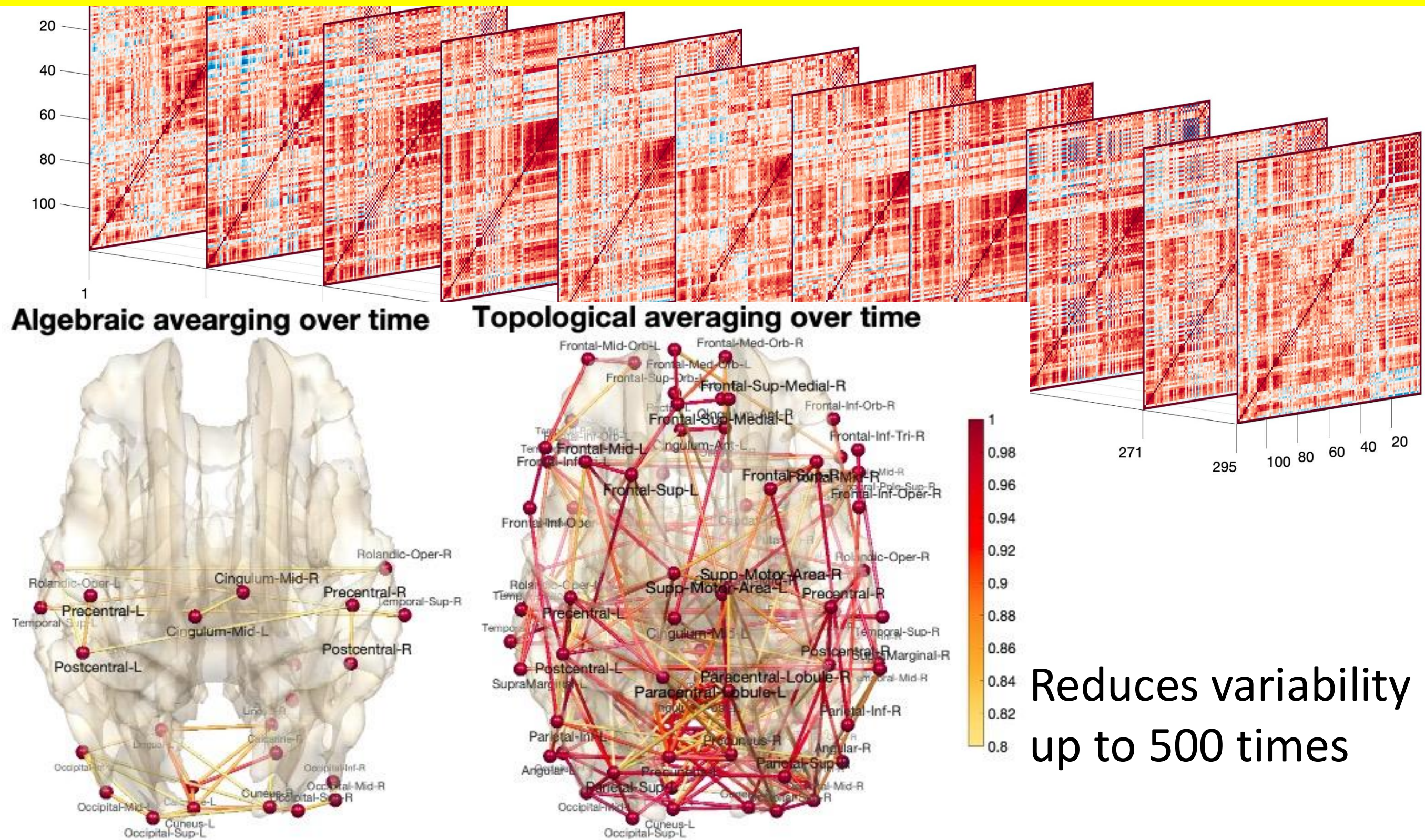
Topological mean

= minimizer of topological distance



Average sorted edge weights over MST and none-MST separately and reconstruct $\rightarrow$ minimizer of topological distance

Superiority of topological averaging in rs-fMRI connectivity



Reduces variability
up to 500 times

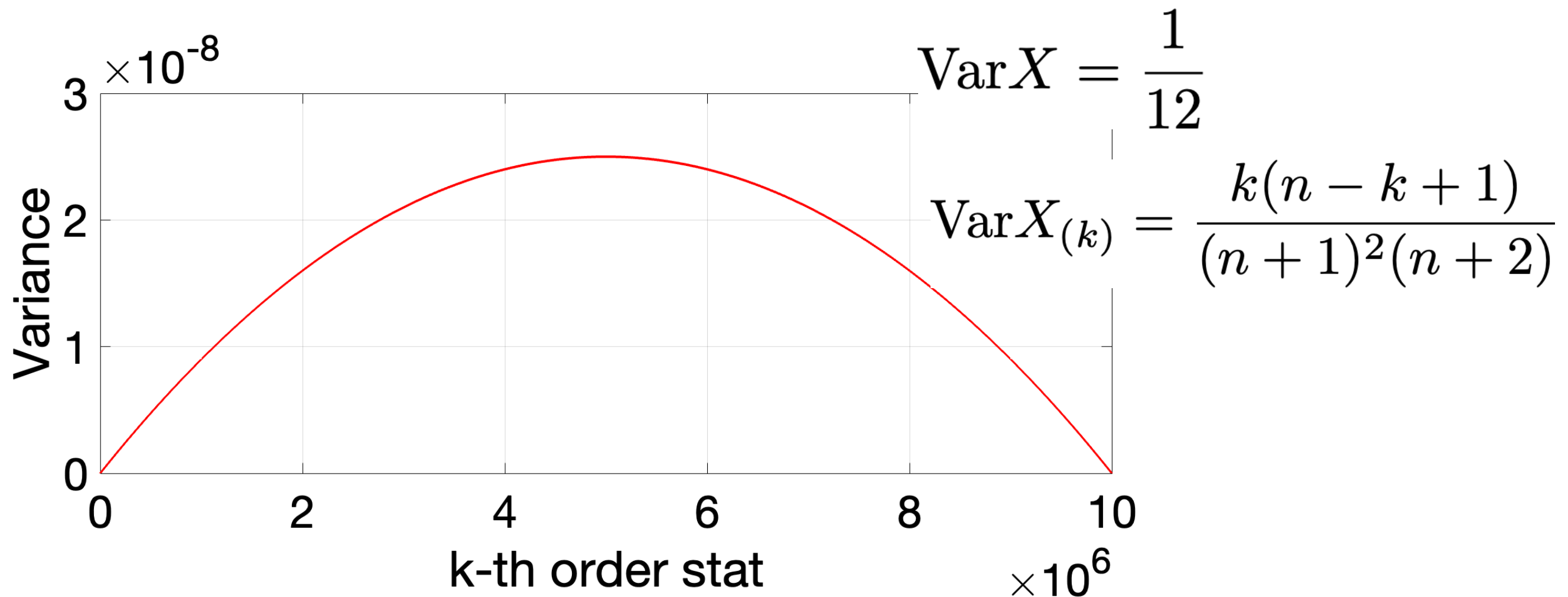
400 subjects

Chung et al. 2024,

Why topological methods reduce variability?

For random variables , $X_1, \dots, X_n \sim \text{Unif}(0,1)$

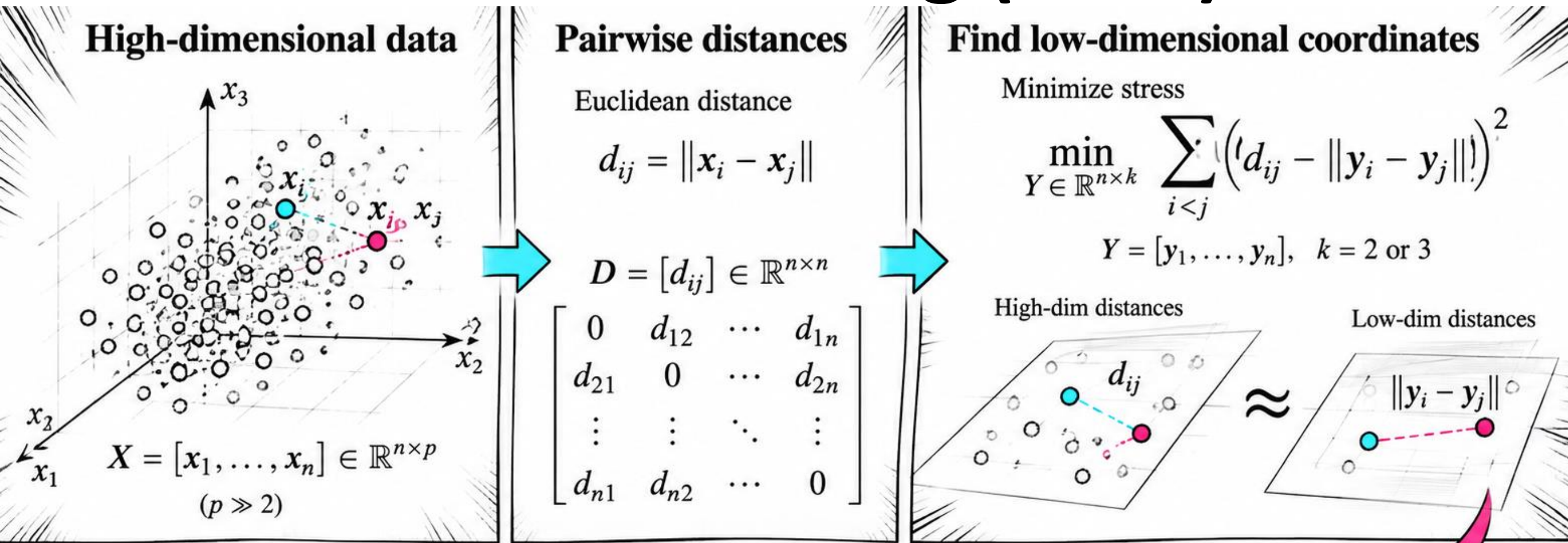
Order statistics $X_{(1)} < X_{(2)} < \dots < X_{(n)}$



Variance reduction $\sim \mathcal{O}(\frac{1}{n})$

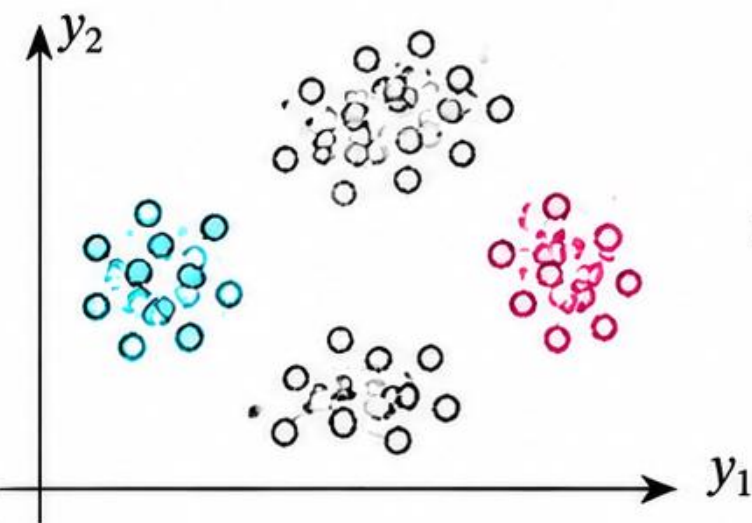
Project: Demonstrate stability of topological features

Multidimensional scaling (MDS)



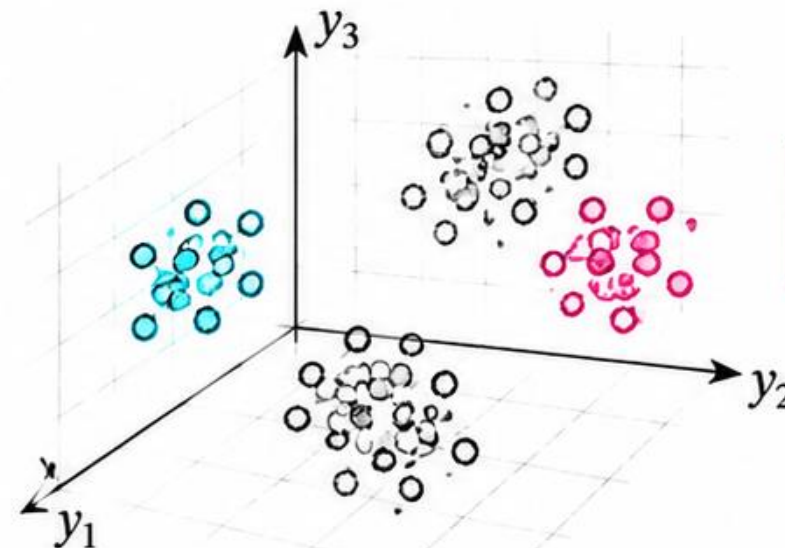
Low-dimensional embedding

$k = 2$ (example)



Geometry
in low dimensions
preserves distances

$k = 3$ (example)



$$\|y_i - y_j\| \approx d_{ij}$$

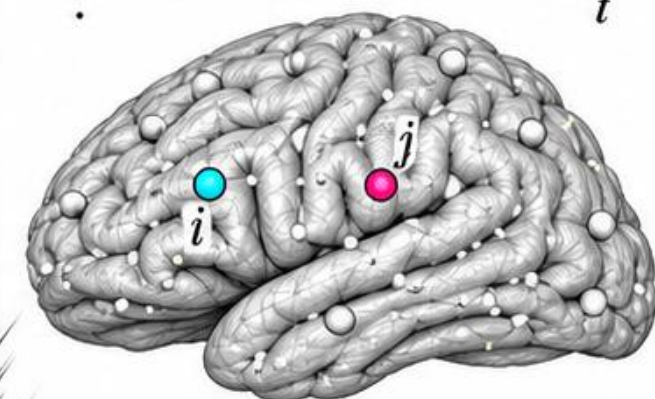
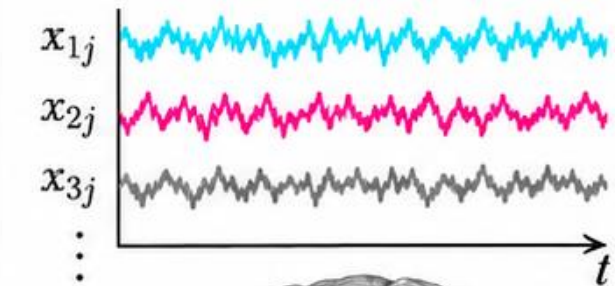
Project: develop MDS on manifolds

Geometry of correlation matrices

1. Measurements at nodes

At each brain region (node) j , we have a time series

$$\mathbf{x}_j = (x_{1j}, \dots, x_{Tj})^\top \in \mathbb{R}^T$$

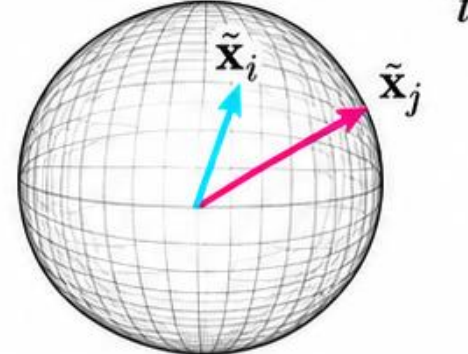
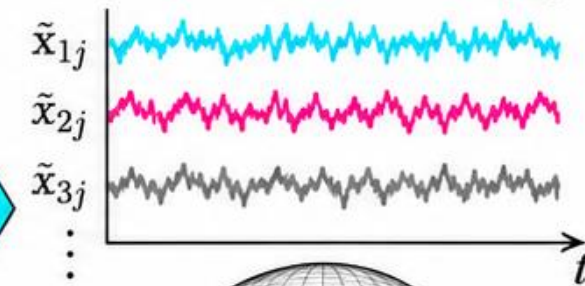


n brain regions (nodes) on the cortical surface

2. Normalize each node

Scale and translate each time series

$$\tilde{\mathbf{x}}_j = \frac{\mathbf{x}_j - \bar{\mathbf{x}}_j}{\|\mathbf{x}_j - \bar{\mathbf{x}}_j\|}, \quad \tilde{x}_{tj} = \frac{1}{T} \sum_{t=1}^T x_{tj}$$

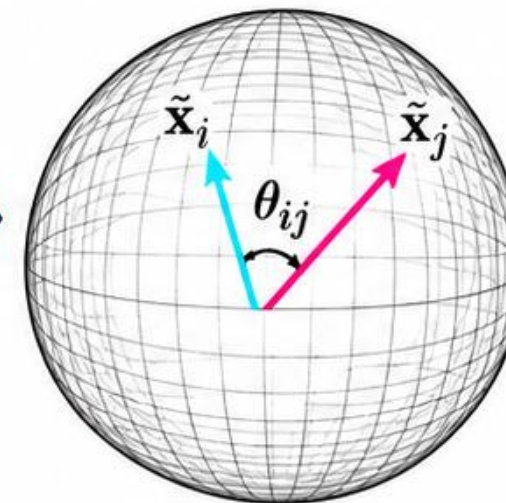


Each node is a point on the unit sphere $S^{T-1} \subset \mathbb{R}^T$

3. Correlation as angle

Pearson correlation between nodes i and j

$$\rho_{ij} = \tilde{\mathbf{x}}_i^\top \tilde{\mathbf{x}}_j = \cos \theta_{ij}$$



$$-1 \leq \rho_{ij} \leq 1$$

$\rho_{ij} = 1$: same direction

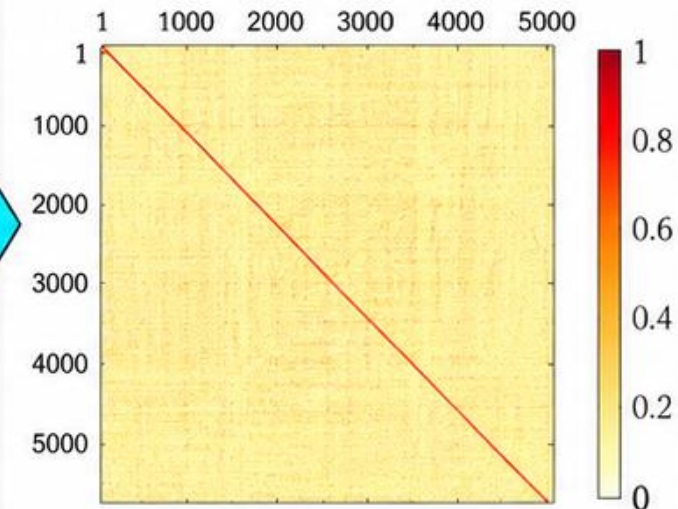
$\rho_{ij} = 0$: orthogonal

$\rho_{ij} = -1$: opposite direction

4. Correlation matrix and network

Collect all pairwise correlations

$$\mathbf{R} = (\rho_{ij}) \in \mathbb{R}^{n \times n}$$



$\mathbf{R}$ is symmetric, positive semidefinite, with ones on the diagonal.

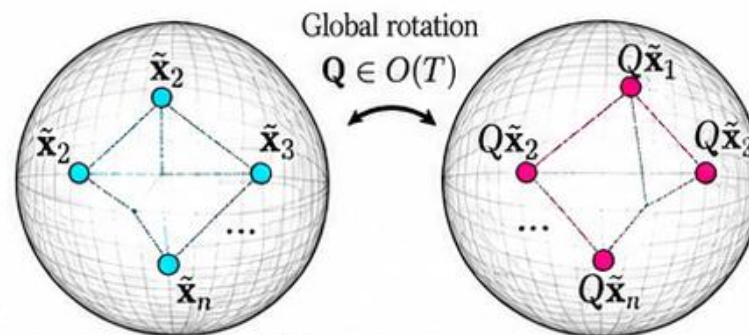
5. Geometry: correlation matrices are Gram matrices

Let $\mathbf{X} = [\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n] \in \mathbb{R}^{T \times n}$ (columns are unit vectors)

$$\mathbf{R} = \mathbf{X}^\top \mathbf{X}$$

$$R_{ij} = \tilde{\mathbf{x}}_i^\top \tilde{\mathbf{x}}_j = \cos \theta_{ij}$$

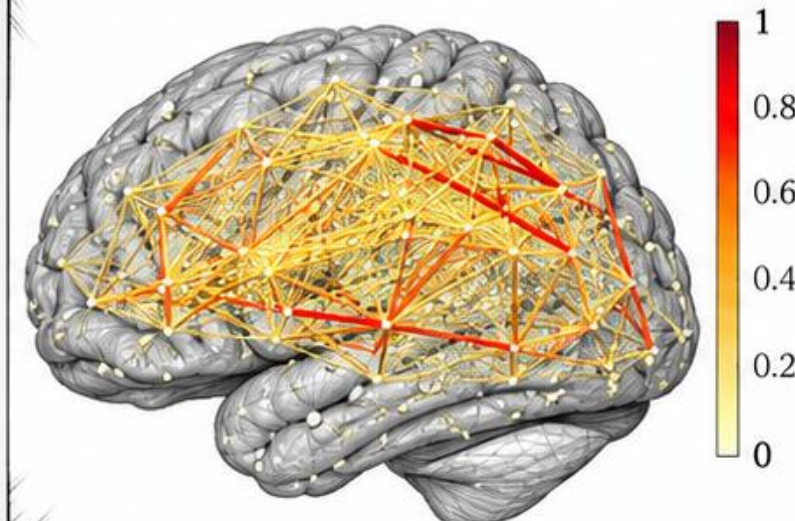
$\mathbf{R}$ depends only on the relative angles between the n unit vectors, not on their global rotation.



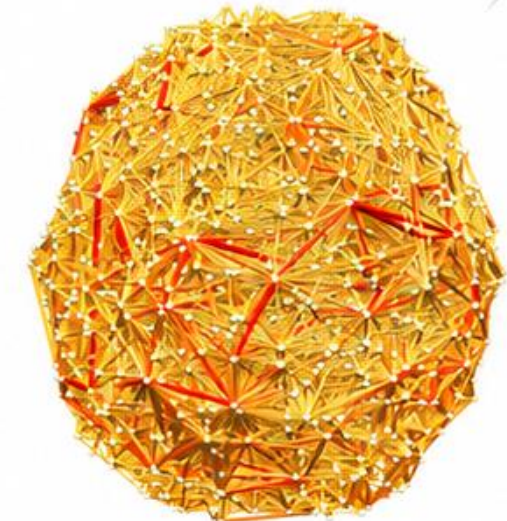
Configuration on S^{T-1}

$$\mathbf{X} \rightarrow \mathbf{QX} \implies \mathbf{R} = \mathbf{X}^\top \mathbf{X} \text{ (unchanged)}$$

6. Example: brain network from correlation matrix

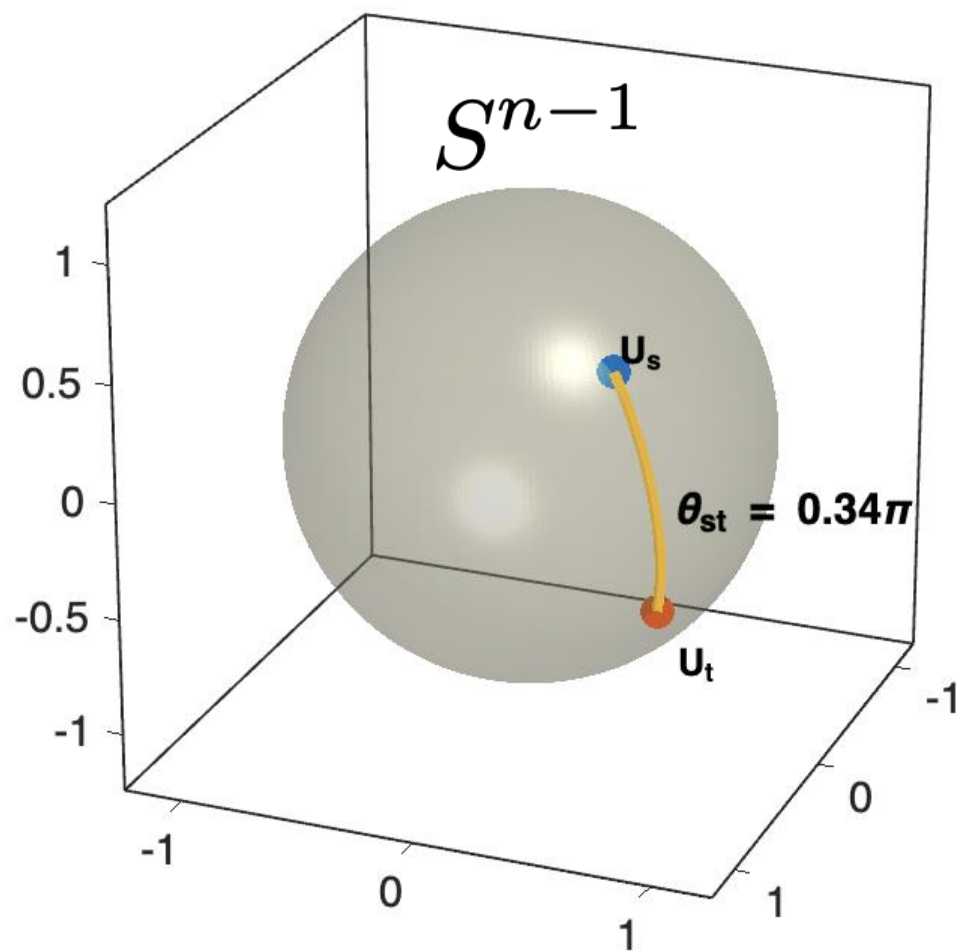


Edges show strong positive correlations (thresholded)

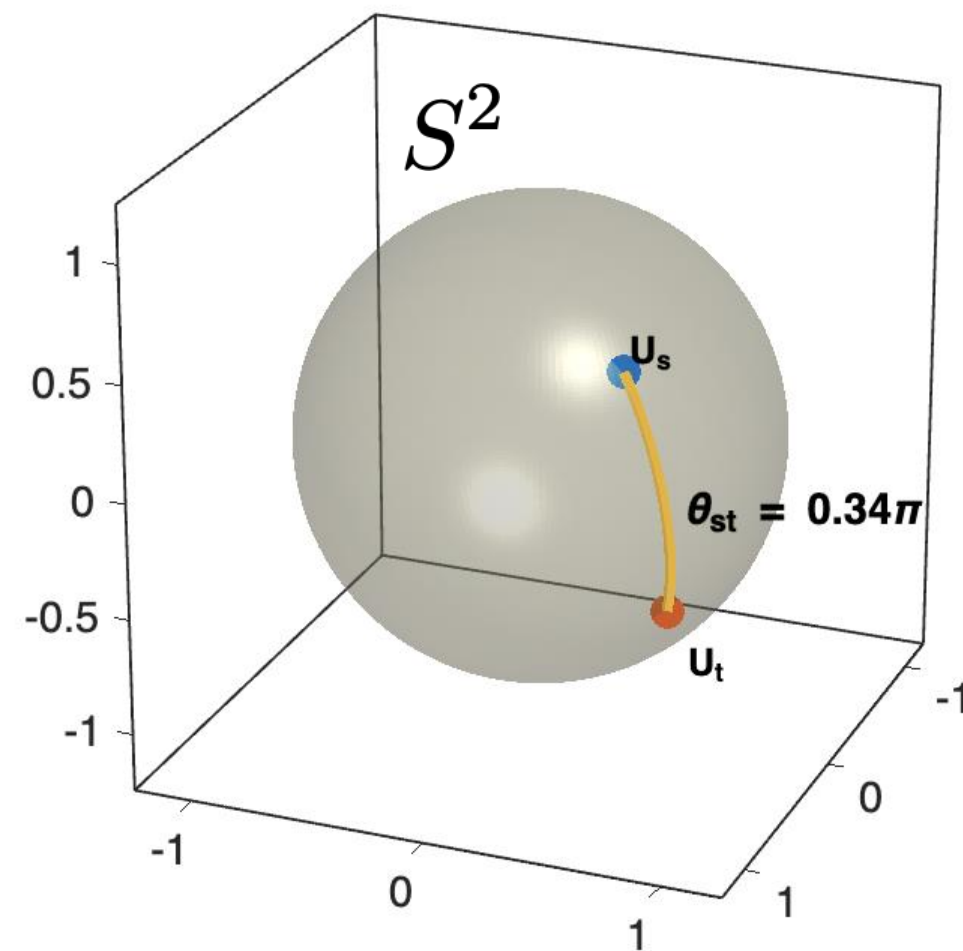


Same network viewed from above

Multidimensional scaling from hypersphere to two-sphere



Multidimensional
scaling



$$\min_{\mathbf{y}_1, \dots, \mathbf{y}_p \in S^2} \sum_{i,j=1}^p \left[\cos^{-1} \rho_{ij} - d(\mathbf{y}_i, \mathbf{y}_j) \right]^2$$

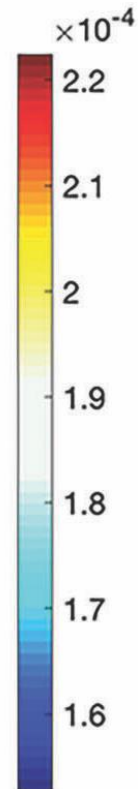
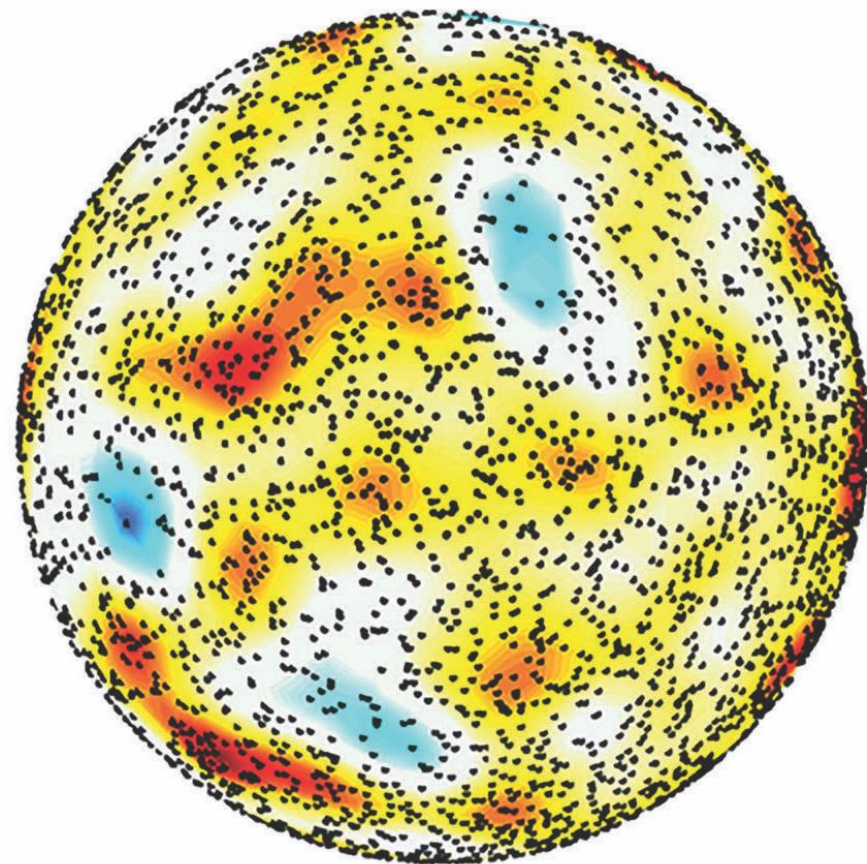


Gradient descent

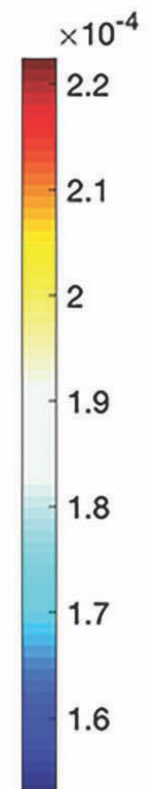
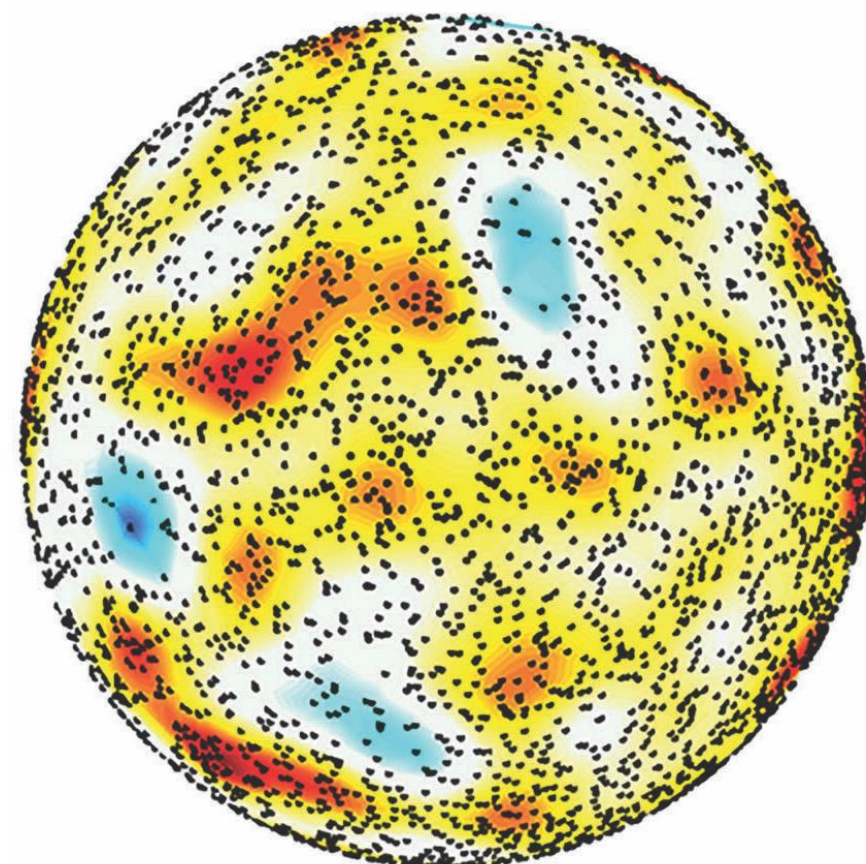
Multidimensional scaling

Chung and Chen 2022 arXiv: 2204.03653

$(n-1)$ -sphere



2-sphere



$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$$



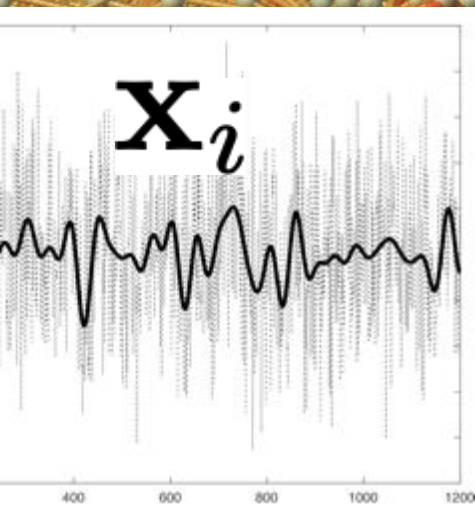
$$\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_p]$$

$$\min_{\mathbf{y}_1, \dots, \mathbf{y}_p \in S^2} \sum_{i,j=1}^p \left[\cos^{-1} \rho_{ij} - d(\mathbf{y}_i, \mathbf{y}_j) \right]^2$$



Gradient descent

Spherical embedding of correlation network



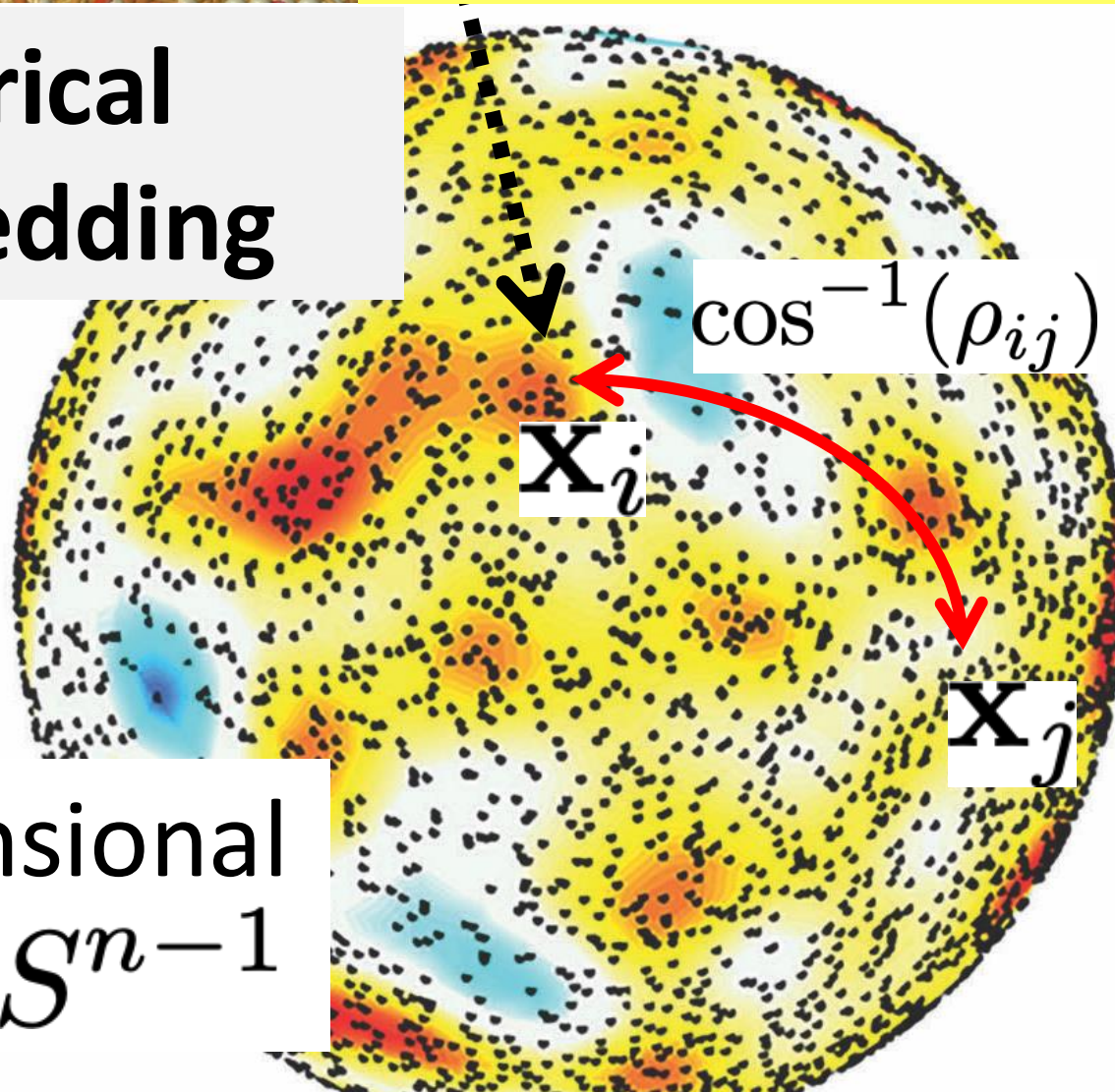
fMRI time series $\mathbf{X}_i$ at voxel i

Translate and scale:

$$\|\mathbf{x}_i\|^2 = \mathbf{x}_i^\top \mathbf{x}_i = 1, \quad \sum_{i=1}^p x_{ij} = 0$$

Spherical
embedding

n -dimensional
sphere S^{n-1}



Correlation

$$\rho_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$$

2

1.9

1.8

1.7

Chung and Chen 2022

arXiv: 2204.03653

$$\min_{\mathbf{Y}} \mathcal{L}(\mathbf{X}, \mathbf{Y}) = \sum_{i,j=1}^p \left[\cos^{-1}(\mathbf{x}_i^\top \mathbf{x}_j) - \cos^{-1}(\mathbf{y}_i^\top \mathbf{y}_j) \right]^2$$

Approximately solve
by the Taylor expansion

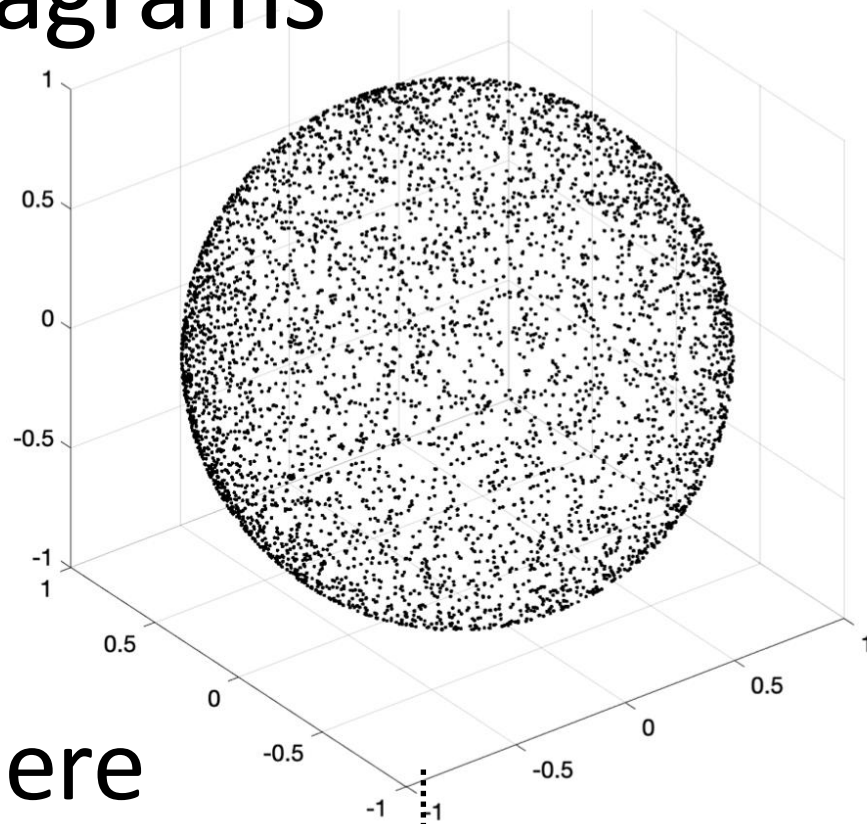
Low rank approximation problem

$$\mathcal{L}(\mathbf{X}, \mathbf{Y}) = \text{tr}(\mathbf{X}^\top \mathbf{X} - \mathbf{Y}^\top \mathbf{Y})^2 = \|\mathbf{X}^\top \mathbf{X} - \mathbf{Y}^\top \mathbf{Y}\|_F^2$$

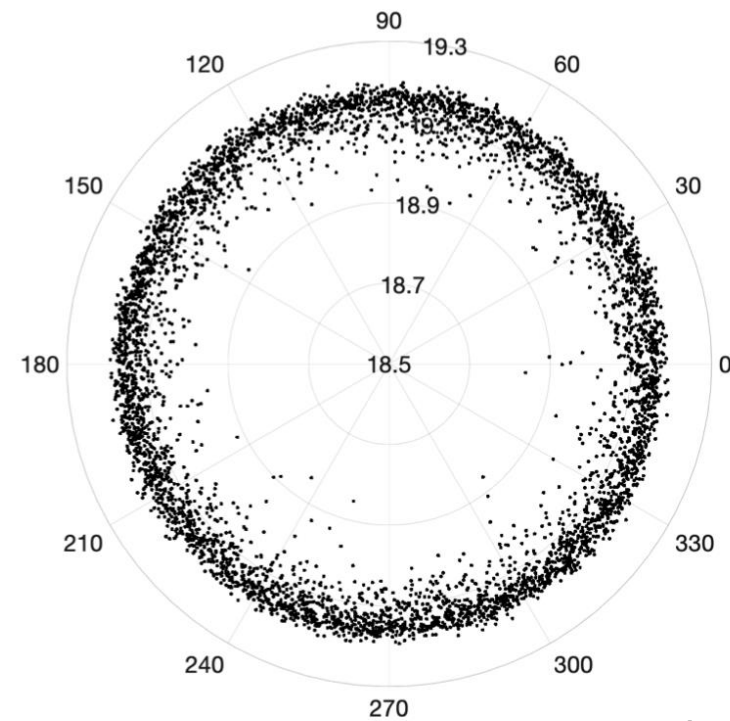
rank $\mathbf{Y} > \text{rank } \mathbf{X}$

Solution: Match 3 principle eigenvectors

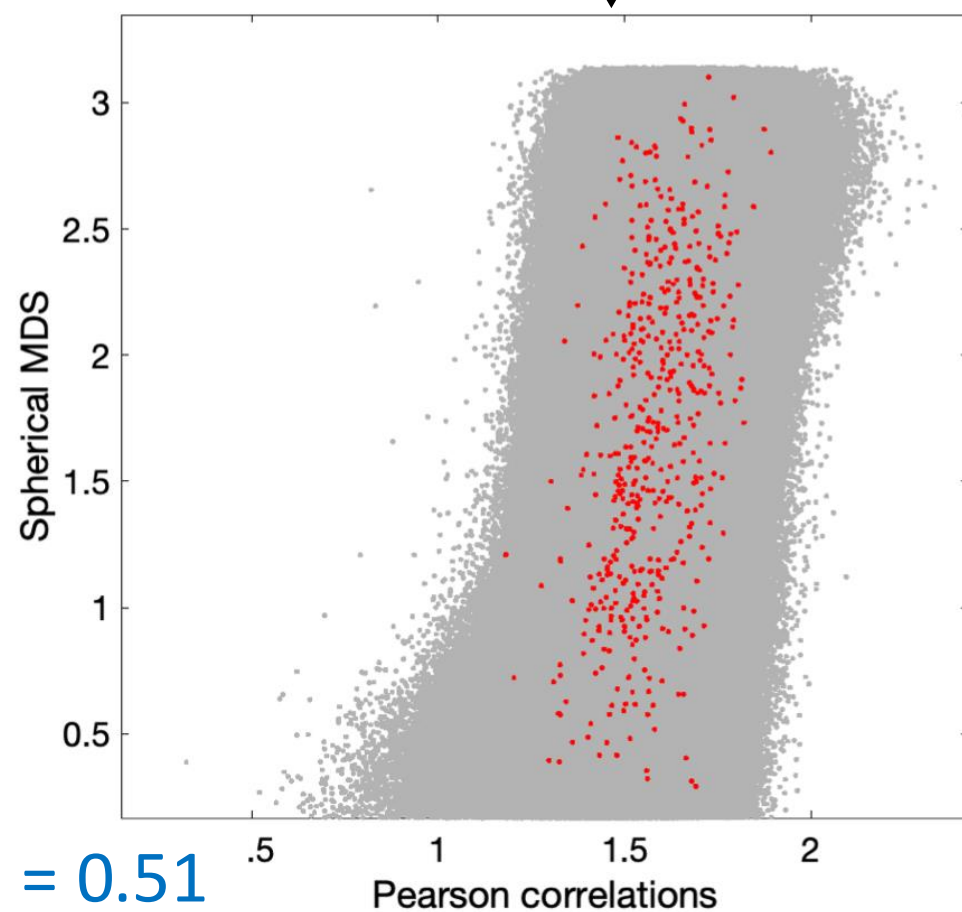
Shepard diagrams



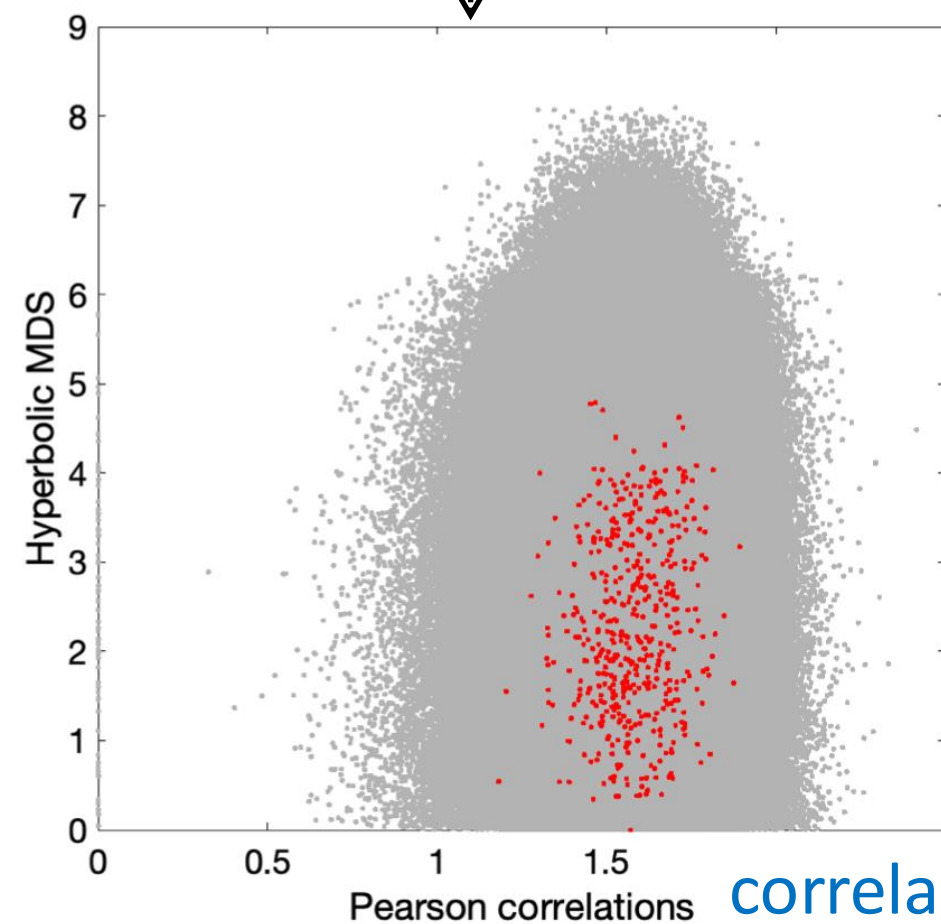
Sphere



Poincare disk

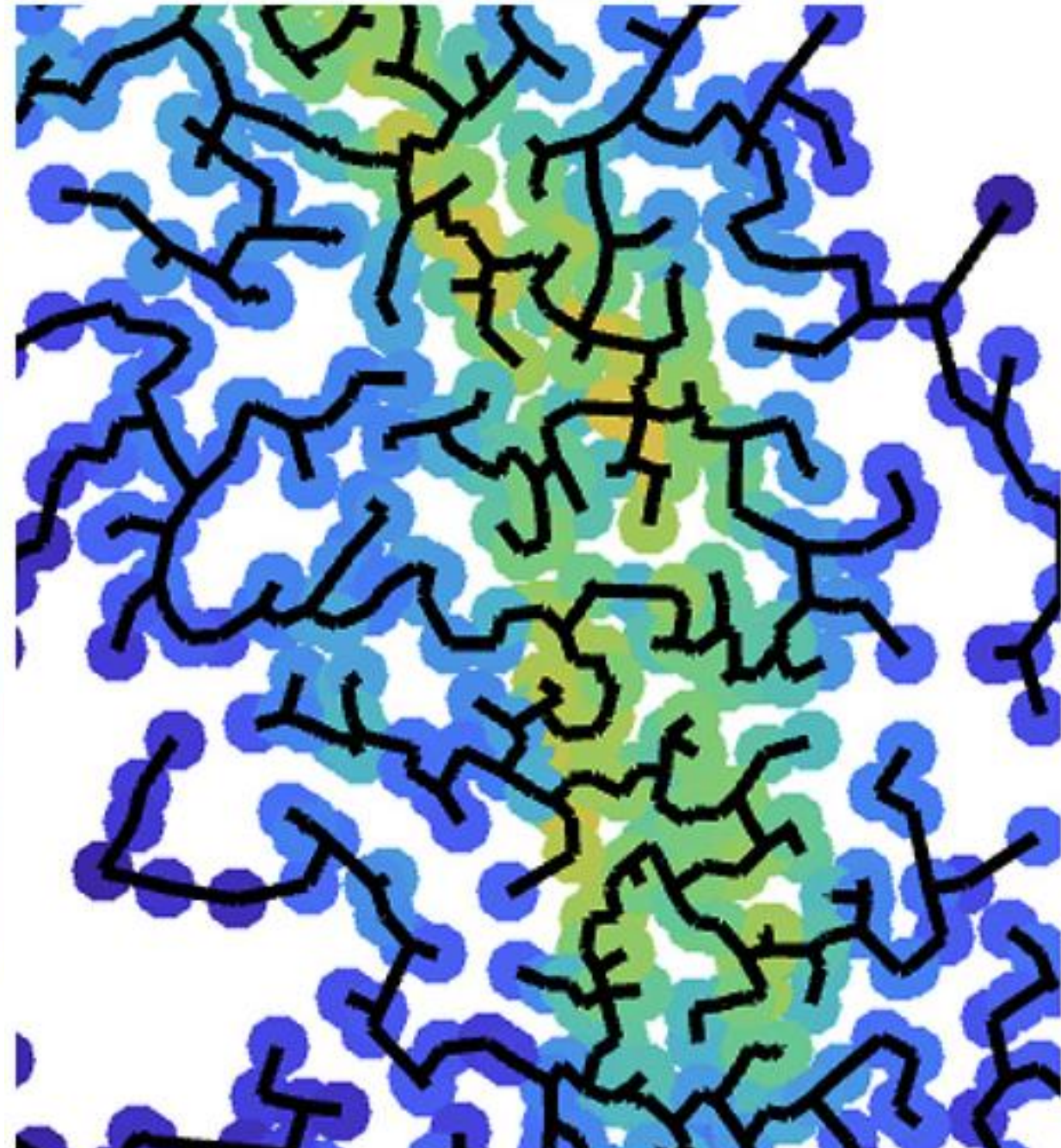
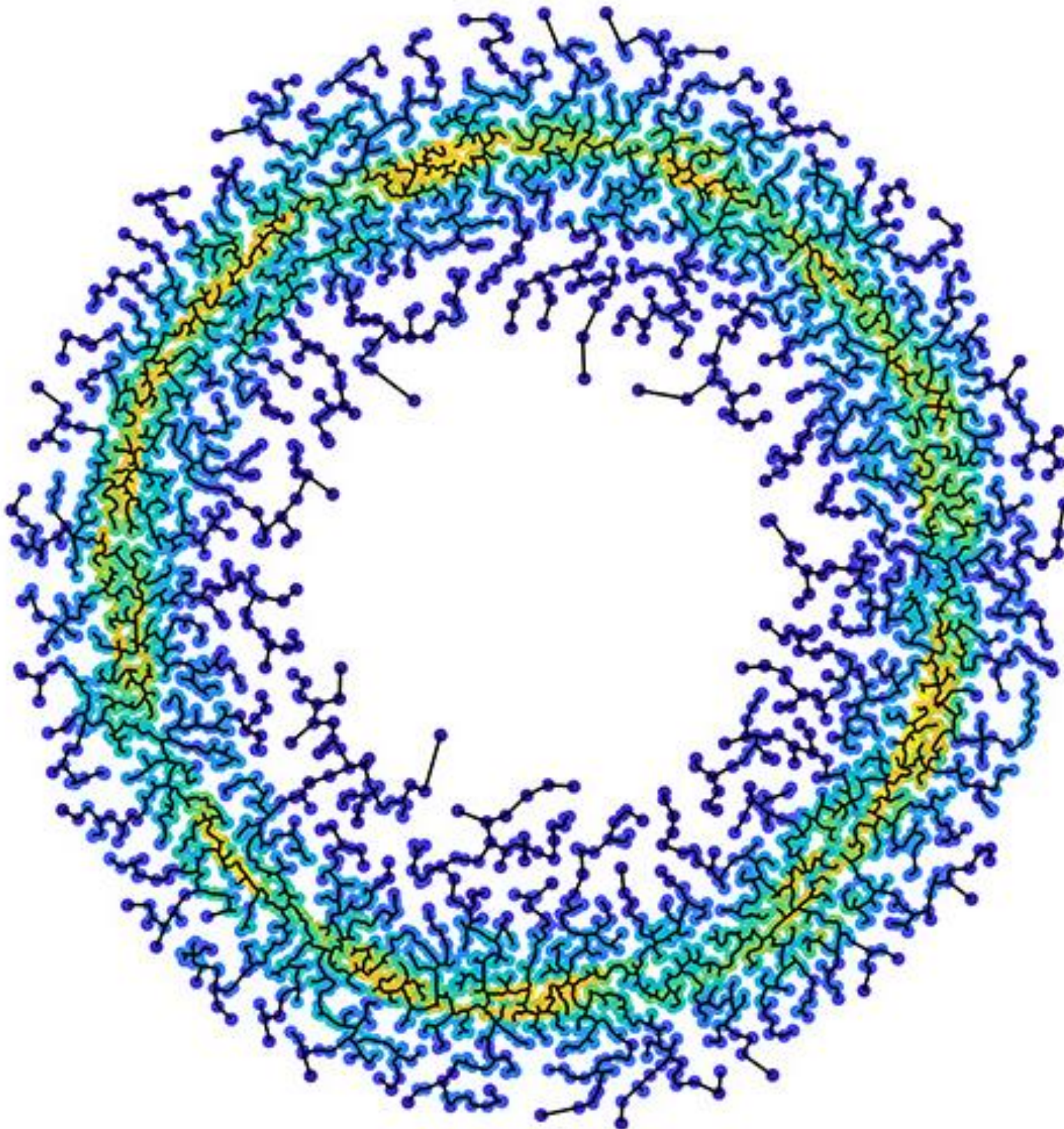


correlation = 0.51



correlation = 0.0018

0D topology (MST) of rs-fMRI correlation network embedded onto **Poincaré disk**:
hyperbolic space with negative curvature



Learning Outcome

1. Differentiate geometric and topological distance.
2. Can average by minimization
3. Can compute Wasserstein distance on topological features